

Verify Claims, Not Scores: Evidence-Based Verification of Modular Agents

Ali Atiah Alzahrani[†]

Abstract

When developers change one component of an agent, such as its controller, a learned model or its verifier, they usually judge the change by an aggregate task score. That score cannot tell whether improvement was attainable, which component lost value, or what the agent’s own checks certify. We introduce a *claim-specific verification audit* for modular agents that plan, act, check and refine. Instead of scoring the agent, the audit scores the *evidence*: each conclusion is recorded with the evidence behind it, one of four verdicts (supported, unsupported, unresolved or not evaluated) and the boundary within which it holds. Three tools supply that evidence. Oracle policies measure attainable improvement under an explicitly stated action set, so that a low value can be traced to the evaluation rather than to the environment. Replacing one component at a time with a perfect counterpart locates lost value, with null results read as unresolved whenever a downstream component could mask them. A separate test asks whether the verifier’s score identifies the quantity it is read as bounding. Applied to a constrained portfolio-allocation agent in a synthetic market with known hidden regimes, the audit shows that the value of perfect regime information depends on the action set used to measure it, that the scenario generator discards most of the regime signal while better local fidelity does not improve decisions, and that the runtime verifier can be bypassed with no visible change in outcomes. The contribution is the protocol and the evidential distinctions it enforces; the empirical findings are specific to the agent and environment studied.

1 Introduction

Agent systems are usually improved by changing one component, such as the controller, a tool, a memory, a learned model or a verifier, and then comparing an aggregate task score before and after (5, 4). How much such a score can establish is widely questioned. Benchmark scores often stand in for broader constructs than they measure (1, 2), agent evaluation harnesses contain errors that change conclusions (6, 7), and small differences are often indistinguishable from seed variation (3). We ask a narrower question: given a change to one component, what evidence would *verify* that it helped, *locate* where value was lost if it did not, and *delimit* what each verification signal can support?

An aggregate comparison is uninformative in at least three situations, each documented in a separate literature:

- **The evaluation cannot express the improvement.** The value of information depends on the actions available to exploit it (21), so an evaluation with a restricted action set can report little headroom where much exists.
- **A downstream component masks an upstream one.** A model can improve its own fidelity metric without improving the decisions it feeds (41, 40), and a better upstream module changes nothing if a downstream module ignores its output.

[†]Corresponding author: aliatiah100@gmail.com. The views expressed are those of the author and do not reflect the views of any other individual or entity. This material is for research purposes only and does not constitute investment advice.

- **The verifier does not measure what it certifies.** A score may not identify the target quantity (29), optimising against a proxy check can satisfy it without the intended property (9, 10), and agents that re-sample until a check passes are selecting on the check (67, 66).

In each case the score is consistent with several incompatible explanations. Telling them apart requires the true hidden state, the best decision attainable given that state, and the ability to replace a component with a perfect version of itself. We therefore work in a controlled setting that provides all three: a synthetic market with recorded hidden regimes, in which an agent allocates a portfolio under explicit constraints. The agent contains no language model, and our claims concern modular decision agents of this kind, although the questions the audit asks are not specific to finance.

Approach and what is new. We propose a *claim-specific verification audit*. Its individual tools (oracle policies, component replacement, factorial attribution, conformal calibration) are established. What is new is the object being verified: not the agent’s score, but each *verification claim*, recorded as claim, evidence, verdict and boundary. Making the claim the unit of verification has three consequences that, to our knowledge, have not been made explicit together for agent development:

1. **Headroom is a property of the harness.** Measuring oracle value at fixed information across a lattice of action sets turns a null result into a testable statement about the evaluation, which we call *verifier expressiveness*, rather than a statement about the environment (§4.2).
2. **Null interventions are non-identifying by default.** Component replacements are read as *recovery*, not ablation: a positive result localises an observed bottleneck, whereas a null result is recorded as *unresolved* whenever a downstream module could mask it (§4.3).
3. **The verifier is itself a verification target.** The audit separates sampling error, model misspecification and *score identification*, meaning whether a check measures what it is read as bounding, and keeps decision quality, calibration, admissibility and provenance as distinct claims (§4.4). This exposes a failure that neither outcome metrics nor feasibility and provenance records detect (§6).

Case study. The system under audit, ARC-AGENT, is a reference agent that we built for this study. It is the object of the audit rather than a proposed allocation method, and we make no performance claim for it (§3; App. P). The audit finds that:

- perfect regime information has a small, statistically unresolved value under a two-portfolio evaluation and a clearly positive value under regime-specific weights (§5.1);
- the scenario generator keeps less than a third of the regime signal it receives, and improving its local fidelity does not detectably improve decisions (§5.2, §5.3);
- a joint repair of two generator outputs helps, while their interaction remains unresolved (§5.4);
- bypassing the runtime verifier voids the decision-quality claim with no visible effect on outcomes (§6).

2 Related work

Evaluating agents versus verifying a change. Benchmarking performance, verifying that a *particular* change helped, and diagnosing *why* value was lost are different activities; an aggregate score mainly serves the first (5, 4). Construct-validity accounts ask what an operationalisation measures (1, 2); we ask the same of a *diagnostic* protocol. Related work repairs evaluation harnesses (6, 7), measures gaps between visible and held-out checks (8), studies reward hacking (9, 10), and monitors agents at runtime (13, 11). Because seed counts and pairing decide which conclusions survive (3), every result here states its seeds, pairing and status (Table 14).

Uncertainty a verification signal can address. Separating uncertainty that more data can reduce from uncertainty it cannot is standard (15, 16), and imprecise-probability and partial-identification traditions (20, 19, 18) formalise the set-valued claims our verdicts take. Distribution-free methods (64, 42, 32) control sampling error and the *consequences* of misspecification; we add the prior question of whether a score *identifies* the quantity it is read as bounding (§4.4).

Established tools, a new object. The value of information depends on the actions that can use it (21), and decision-focused learning shows that fidelity and decision quality come apart (41, 40, 30).

Table 1: **What each practice is typically designed to establish.** Entries describe the usual focus of a family of methods, not the limits of every work within it. ✓: addressed by design; “partly”: addressed for a specific case; –: not the focus.

Practice	Question it answers	Null traced to harness	Masking-aware localisation	Quality / admiss. / provenance kept apart	Verifier score identification
Aggregate benchmarking	How well does the agent do?	–	–	–	–
Harness auditing	Is the grader correct?	partly	–	–	–
Component ablation	How much does removal hurt?	–	–	–	–
Runtime verification	Does a run meet a spec?	–	–	partly	–
Decision-focused learning	Does fidelity serve decisions?	–	partly	–	–
Claim-specific audit	Which claims does the evidence support?	✓	✓	✓	✓

Oracles, component replacement, factorial designs and Shapley attribution (26, 27) are all established, and our information–action lattice is elementary as mathematics. Table 1 summarises what we add: a protocol that binds each tool to a named claim, an identification assumption, a verdict and a boundary.

3 The system and environment under audit

Before presenting the audit, we describe the concrete system it is applied to. Every component and signal referred to later is introduced here.

Environment. We use U3, a synthetic benchmark (App. N). Ten assets, three of them defensive, have daily returns whose distribution depends on a hidden market *regime*: bull, stagnation or crisis. The regime switches according to a Markov chain. Crises are rare (about 5% of days), last about 45 trading days on average, and are fat-tailed, with risky assets falling together. Because the simulator records the true regime, three things become possible that are impossible on historical data: regime inference can be scored directly, policies that know the regime can be computed exactly, and any component can be replaced by an exact counterpart. U3 is verification infrastructure, not evidence about financial markets. Each run (seed) simulates a 756-day warm-up followed by 69 monthly decisions, with all estimation strictly walk-forward.

Decision problem and metric. At each rebalance t the agent chooses weights w_t that are long-only and fully invested (the *budget*), lie in a per-asset *box* $[0, 0.25]$, and move at most 0.20 in ℓ_1 norm from w_{t-1} (the *turnover cap*), with 8 bps costs. The metric is the Sharpe ratio, mean return divided by volatility. Because Sharpe can be bought with deeper drawdowns, we compare at matched risk where possible, against a *static frontier*: the best Sharpe of fixed equal-weight/minimum-variance blends at the same maximum drawdown (MaxDD).

The agent under audit. ARC-AGENT makes each decision in five steps (Fig. 1):

1. a *regime detector*, a hidden Markov model (HMM), outputs a posterior over regimes;
2. a *controller*, a transparent rule plus a contextual bandit, sets configuration parameters such as the tail weight and risk level, but never weights or constraints;
3. a *scenario generator*, a regime-conditioned diffusion model, draws $N=512$ scenarios of next-period returns;
4. a *runtime gate*, calibrated by adaptive conformal inference on realised losses, accepts or rejects the scenario set; a rejected set is regenerated up to $B=3$ times, then replaced by a block-bootstrap fallback;
5. an *allocator* solves a convex program trading expected return against variance and conditional value-at-risk (CVaR_α , the mean loss in the worst 5% of scenarios) under the constraints; its output w^{sol} is clipped and renormalised into the committed action w_t .

App. P gives the equations and the full loop (Algorithm 1).

Trust boundary and signals. Adaptive components (the controller and layer A in Fig. 1) may be wrong or adversarial. They can degrade decision quality but cannot enlarge the feasible set,

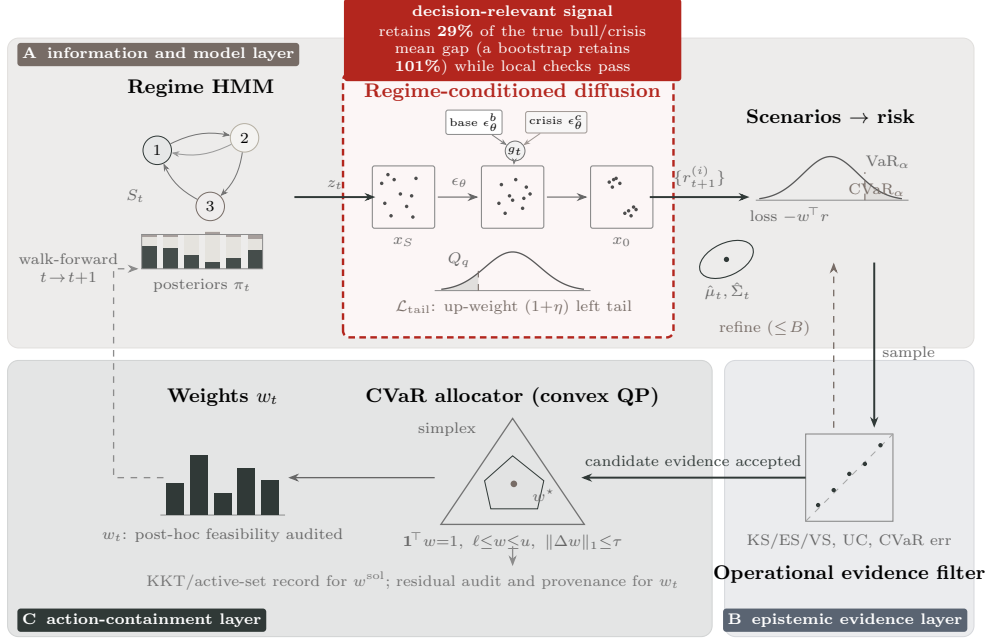


Figure 1: **ARC-AGENT, the agent under audit, and its three verification layers.** Arrows trace one rebalancing decision at date t . **(A) Information and model layer:** a regime HMM turns past returns into regime posteriors π_t (walk-forward), and a regime-conditioned diffusion generator, whose base and crisis experts are mixed by a gate g_t and trained with a tail-weighted loss $\mathcal{L}_{\text{tail}}$, draws return scenarios from which the risk quantities (VaR_α , CVaR_α) and moments ($\hat{\mu}_t, \hat{\Sigma}_t$) are computed. The red dashed box marks where the audit finds decision-relevant signal lost: the generator retains 29% of the true bull–crisis mean gap (a bootstrap retains 101%) while local checks pass (§5.2). **(B) Epistemic evidence layer:** the operational evidence filter scores each scenario set with calibration checks (KS, energy and variogram scores, unconditional coverage, CVaR error) and either accepts it or requests refinement, at most B times; it certifies no CVaR_α bound (§4.4). **(C) Action-containment layer:** the CVaR allocator solves a convex QP under the budget, box and turnover constraints. Its KKT and active-set record is optimality evidence for the solver output w^{sol} , whereas the committed weights w_t receive only a post-hoc feasibility audit, which supports admissibility for the reported runs.

because the constraint parameters (ℓ, u, τ) and the budget B belong to a trusted validator in the action-containment layer (C). The system emits signals bearing on four distinct properties: *decision quality* (is w_t a good risk-adjusted choice?), *calibration* (do scenarios match outcomes? monitored by the gate), *admissibility* (does w_t satisfy the constraints? checked by a post-hoc audit) and *provenance* (can the solver output be traced to a logged optimisation? recorded by the KKT optimality conditions).

Protocol. A *paired* contrast compares two arms on identical simulated paths (common random numbers). Unless marked otherwise, intervals are 95% bootstrap intervals over seeds. Experiments with 10–24 paired seeds are *confirmatory* and those with 5–6 seeds *exploratory* (Table 14). Preregistration is author-reported, with one disclosed amendment (App. T).

4 A claim-specific verification audit

The audit takes a proposed conclusion, such as “the generator is the bottleneck”, and asks whether the evaluation can *express* the improvement it concerns, whether the evidence *identifies* that improvement, and where the inference *stops*. For ARC-AGENT we pose four questions (evidence and permitted claims in Table 4, App. B):

- Q1** Was improvement attainable under an action set able to express it?
- Q2** Which component replacement recovers downstream value?

- Q3** Does improved local fidelity verify downstream improvement?
Q4 Do joint repairs reveal structure that one-at-a-time checks miss?

4.1 Verification claims and verdicts

A *verification claim* is a conclusion recorded as $\text{Claim} \rightarrow \text{Evidence} \rightarrow \text{Verdict} \rightarrow \text{Boundary}$, with one of four verdicts:

- *Supported*: the evidence supports the claim, which may be relied on only inside its boundary.
- *Unsupported*: the evidence conflicts with the claim, or a required identification condition is unmet; the claim must not be used for selection or certification.
- *Unresolved*: the experiment cannot distinguish the relevant explanations; the claim needs more evidence, not clearance. It is not a weak form of *supported*.
- *Not evaluated*: no experiment was run.

Table 2 is the resulting ledger for ARC-AGENT; App. L expands each boundary.

4.2 Oracle headroom: can the evaluation express the improvement?

An *oracle* is a policy with perfect knowledge of the hidden state. It obeys the system’s own constraints and costs and differs from the system only in what it knows and which actions it may take. Indexing oracles by an information set I and an action set A gives a lattice of attainable values,

$$J^*(I, A) = \sup_{\pi \in \Pi(I, A)} J(\pi), \quad (1)$$

where $\Pi(I, A)$ is the set of policies that use only information in I and act in A , and J is the metric. J^* is monotone in both arguments (Prop. 1). A headroom estimate is therefore a property of an (I, A, J) cell, not of the information alone. A low estimate in one cell cannot show that the information has little value; if headroom rises when A widens at fixed I , the low estimate came from the evaluation.

For ARC-AGENT we fix I to the true regime path and use four action sets of increasing expressiveness: **L0**, the best static equal-weight/minimum-variance blend, with no regime knowledge; **L1**, switching between those two portfolios with perfect regime knowledge; **L2**, a separate weight vector $w^*(k)$ for each regime k ; and **L3**, the ex-post optimal weight path.

4.3 Oracle-replacement interventions: where is value lost?

We replace one component at a time with an oracle counterpart and re-run the system. Every arm receives the true regime, so regime detection is never the confound. The *perception* replacement feeds the true regime to the generator; the *full generator bypass* hands the allocator scenarios drawn from clean, regime-conditional history.

The evidence is asymmetric. A replacement that *recovers* value identifies an observed bottleneck. A replacement that recovers nothing leaves open whether the component matters, because a downstream module that ignores its input masks any upstream repair. Such a null is recorded as *unresolved*, not *unsupported*.

Some repairs act only jointly. The generator reaches the allocator through three channels: the scenario set itself (tail shape and dependence), the conditional mean $\hat{\mu}$ and the covariance $\hat{\Sigma}$. A complete 2^3 factorial switches each channel independently between the generator and clean history. With $v(S)$ the paired Sharpe gain when the channels in S are repaired, we summarise the table with Shapley values ϕ_i and Shapley–Taylor interaction indices ϕ_{ij} (26, 27), and call $\sum_{i < j} \phi_{ij}/v(N)$ the *interaction share*. These summaries are descriptive and non-additive (App. U).

4.4 What can the runtime gate certify?

Scenarios come from the generator’s model p_θ , not from the true outcome law $\mathbb{P}$, so the error in the allocator’s CVaR estimate has two parts: a *sampling* term, which shrinks as N grows, and a *misspecification* term ($p_\theta \neq \mathbb{P}$), which does not (App. W). A third question sits above both: does the verification score *identify* the quantity it is read as bounding?

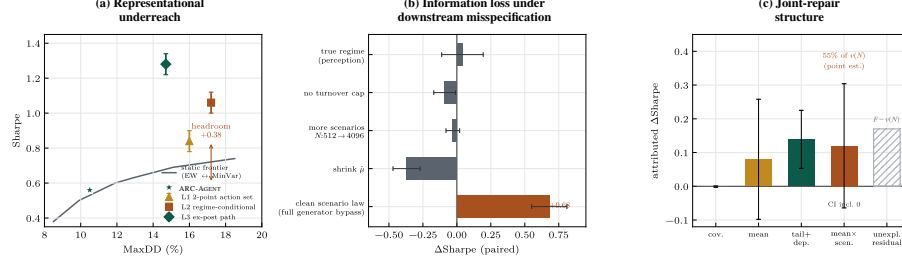


Figure 2: **Three failure modes exposed by the audit** (Sharpe; seeds and intervals in Table 14). **(a) Representational underreach:** regime-conditional weights (L2) sit $+0.38$ above the static frontier at matched drawdown, but switching between two portfolios (L1) only $+0.18$, a paired gain over L0 of $+0.07 \pm 0.08$. **(b) Interventional diagnosis:** the true-regime replacement recovers nothing and is *unresolved* because the generator can mask it; removing the turnover cap, adding scenarios and shrinking the mean also fail; only the full generator bypass recovers the gap ($+0.68$). **(c) Joint-repair structure** (exploratory; Sharpe only): the mean-plus-scenario arm improves over baseline; the 55% interaction share is a point estimate whose six-seed interval includes zero.

The gate is calibrated on realised returns, the only direct evidence about misspecification, and acts as an *operational evidence filter*. Its score compares one realised loss with a scenario CVaR. CVaR is not elicitable (29), so this score does not identify CVaR_α (condition C5); a joint VaR–CVaR score would (28). In addition, refine-until-pass selects on the calibration statistic, which a union bound corrects only for bounded B and only if each attempt is valid on its own (condition C6; App. W). The gate is therefore not a decision-quality certificate: that claim is *unsupported*, and no empirical result in §5 depends on the gate.

4.5 Trusted action boundary and attack surfaces

Controller-settable parameters are *untrusted*: they can degrade quality but not enlarge the feasible set. Constraint parameters and B are *trusted* and validator-owned (App. O; Prop. 3). We specify five attacks: **A1** randomises controller outputs within their ranges; **A2** poisons memory by flipping the regime posterior; **A3** bypasses the runtime gate; **A4** is an adversarial controller that maximises turnover; **A5** would compromise the validator. A1–A4 were executed; A5 was not.

5 Verification findings

All numbers are measured and scoped to U3 and the audited architecture. Each subsection answers one question and ends with a verdict.

5.1 Q1: Restricted evaluations can hide attainable headroom

With perfect regime knowledge and identical constraints and costs, the four oracles reach Sharpe 0.78 (L0), 0.84 (L1), **1.06 (L2)** and 1.28 (L3), against 0.56 for ARC-AGENT (Table 10; $rf=1.6\%$, see App. V for $rf=0$).

L1 is an *evaluation rule* with a restricted action set, not a weak baseline. Under it, perfect regime information is worth a paired $+0.07 \pm 0.08$ Sharpe over L0, an interval that spans zero. Widening only the action set raises the paired L2–L0 gain to $+0.28 \pm 0.08$. What limited the measured headroom was the evaluation, not the environment. ARC-AGENT does not capture this headroom (it sits 0.04 below the static frontier at matched drawdown), and no other scenario-based method does either (App. G).

Verdict: supported, for U3 and the stated action sets.

5.2 Q2: Oracle replacements localise an observed generator bottleneck

Giving the generator the true regime instead of the HMM posterior changes nothing (16 seeds, paired: $+0.041 \pm 0.153$), although the HMM is poor (crisis recall falls from 0.54 to 0.07 as tails fatten). This does *not* clear perception: a generator that ignores its conditioning masks any benefit, and the detector $\times$ generator factorial was not run. Removing the turnover cap (-0.09), shrinking $\hat{\mu}$ (-0.37) and raising N from 512 to 4096 (-0.03) also recover nothing; more scenarios reduce sampling error but not misspecification. Only the full generator bypass recovers the gap (Table 5).

The reason is visible in the generator’s output. Given perfect regime conditioning, it reproduces only **29%** of the true bull–crisis difference in conditional means (90 of 313 bps), which we call *mean-gap retention*. A regime-conditional bootstrap, which resamples history from the same regime, reproduces **101%** and scores higher downstream (0.85 versus 0.56). The regime signal reaches the generator, yet no end-to-end score reports its loss.

Verdict: *supported* for the generator as an observed bottleneck under this architecture; *unresolved* for perception.

5.3 Q3: Local fidelity does not verify downstream improvement

Sweeps over diffusion steps and crisis oversampling implicate two contributors to the retention loss: truncated reverse diffusion and under-representation of crises in training. A repair targeting both raises retention from **47%** to **186%** and corrects the direction of the regime mean shift (cosine $0.59 \rightarrow 0.97$). Yet the paired Sharpe change is $+0.075 \pm 0.174$, and the repaired generator (0.63) still trails the bootstrap (0.85). The repair overshoots the size of the shift (208%), so the mean vector’s relative error grows ($0.82 \rightarrow 1.18$; Table 7).

Verdict: *supported*. Better local fidelity does not establish better decisions. The claim is one of non-identification, not of a zero effect.

5.4 Q4: Joint repairs reveal structure missed by one-at-a-time checks

Repairing the mean and the scenarios together improves on the all-generator baseline by $+0.221$ (six-seed paired 95% CI ± 0.160), essentially all that the three channels recover together ($v(N) = +0.218$). Each channel alone does less (mean $+0.020 \pm 0.206$; scenarios $+0.079 \pm 0.066$; covariance nothing). The Shapley–Taylor estimate assigns 55% of $v(N)$ to the mean $\times$ scenario interaction (Table 15), but its six-seed interval (± 0.18) includes zero, and the factorial records Sharpe only (App. D).

Verdict: *supported* for the joint repair improving on baseline; *unresolved* for superadditivity and the size of the interaction.

5.5 Summary: different signals support different claims

Table 2 collects the verdicts. Evidence for one property does not transfer to another: feasibility audits support admissibility for the reported runs only; the KKT record supports provenance for w^{sol} , not w_t ; and because the gate’s score does not identify CVaR_α , the decision-quality claim is *unsupported*.

6 Red-teaming the verifier and its trusted boundary

If decision quality cannot be certified, what can the harness still be trusted to say? Attacks A1, A2 and A4 compromise adaptive components but leave the trusted boundary intact. Under all three, every committed action passed the feasibility audit and kept its provenance record (Table 3). This holds only for attacks that do not touch the validator.

A3 attacks the verifier itself, and nothing notices. Bypassing the gate removes the evidential basis for the decision-quality claim, yet feasibility and provenance stay intact and the outcome stays benign: under A3 the maximum drawdown, 9.9%, is the *lowest* in the table. No outcome metric we report would have flagged the bypass. A verifier can lose its evidential validity without any immediately poor outcome, and feasibility and provenance cannot substitute for it, because they never depended on the gate.

Table 2: **Claim ledger for ARC-AGENT** (boundaries expanded in App. L). *Unresolved* is not a weaker form of *supported*; it marks a claim the evidence cannot settle either way.

Verification claim	Evidence	Verdict	Boundary
Decision-relevant information is present and expressible	L2 information–action hierarchy	supported	U3, specified action set
ARC-AGENT captures it	matched-risk frontier	unsupported	current architecture
Generator is an observed bottleneck	full generator bypass	supported	current trained generator
Improving perception would not improve downstream value	true-regime replacement	<i>unresolved</i>	masked by generator
Joint mean+scenario repair improves on the all-generator baseline	2 ³ coalition table	supported	U3; 6 seeds, exploratory; Sharpe only
Mean×scenario superadditivity	Shapley–Taylor interval	<i>unresolved</i>	interval includes zero at 6 seeds
Executed actions are admissible	post-hoc residual checks	supported	reported runs only; not a universal guarantee
Solver outputs are traceable	KKT and active-set records	supported	applies to w^{sol} ; auditability unmeasured
Implemented gate certifies decision quality (CVaR)	implemented one-step score	unsupported	identification condition (C5) unmet
Trusted-path compromise contained	A5	<i>not evaluated</i>	attack unexecuted

Table 3: **Attack surfaces and what survives each executed attack.** Decision quality, admissibility and provenance are reported separately because they are different claims. Exploratory: 5 seeds, unpaired; levels in Table 11. The feasibility audit is a post-hoc check, not a construction guarantee; solver provenance is a record for w^{sol} , not KKT-optimality of w_t . **A5 was not executed**, so no claim covers compromise of the trusted path.

Attack	Claim surface	Effect on decision-quality evidence	Feas. audit	Solver prov.
A3 gate bypass	quality-verification layer	gate bypassed; metrics benign	passed	retained
A1 corrupted controller	untrusted configuration	degraded, within noise	passed	retained
A2 poisoned memory	untrusted memory	degraded, within noise	passed	retained
A4 adversarial controller	untrusted configuration	moved <i>along</i> frontier	passed	retained
<i>A5 validator compromise: specified but not executed; no robustness claim.</i>				

The preregistered containment hypothesis H3 was *not supported*: under A4, ARC-AGENT’s apparent advantage over a rule-based control (0.83 versus 0.72 Sharpe) comes at a different risk level (16.7% versus 19.4% MaxDD), and at matched risk the attack moved the system *along* the frontier (App. H).

7 Lessons, limitations and conclusion

Lessons for reliable agent development (in full in App. A):

1. Check that the evaluation can express the target improvement before reading anything into a null.
2. An unchanged score fits both a useless and a masked component, so verify updates by intervention.
3. Read interventions asymmetrically: a positive replacement localises a bottleneck; a null one does not show irrelevance.
4. Decision quality, calibration, admissibility and provenance are different claims, and none certifies another.
5. Red-team the verifier as well as the agent: it can be bypassed while outcomes stay benign.

Limitations. Every statement is scoped to U3, one architecture and one generator family (App. L). Perception is unresolved under masking; the interaction results rest on six seeds and Sharpe only; conditions C5 and C6 behind any certificate reading of the gate are unproved, though no empirical result depends on the gate; A5 was not executed; and no language-model controller, model judge or prompt-injection attack was evaluated.

Conclusion. The unit of agent verification should be the claim, not the score. Treating it so turns a flat score into three separate questions: could the change have helped under the evaluation used, where was value lost, and what does each check certify. In the case studied, this exposed a restricted evaluation, an observed generator bottleneck, a gap between local fidelity and decision value, and a verifier that could be bypassed without any visible effect on outcomes.

References

- [1] A. Z. Jacobs and H. Wallach. Measurement and fairness. In *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pp. 375–385, 2021.
- [2] I. D. Raji, E. M. Bender, A. Paullada, E. Denton, and A. Hanna. AI and the everything in the whole wide world benchmark. In *NeurIPS Track on Datasets and Benchmarks*, 2021.
- [3] R. Agarwal, M. Schwarzer, P. S. Castro, A. C. Courville, and M. G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems*, volume 34, pp. 29304–29320, 2021.
- [4] S. Rabanser, S. Kapoor, P. Kirgis, K. Liu, S. Utpala, and A. Narayanan. Towards a science of AI agent reliability. In *ICML*, 2026. arXiv:2602.16666.
- [5] X. Liu, H. Yu, H. Zhang, Y. Xu, X. Lei, H. Lai, Y. Gu, H. Ding, K. Men, K. Yang, S. Zhang, X. Deng, A. Zeng, Z. Du, C. Zhang, S. Shen, T. Zhang, Y. Su, H. Sun, M. Huang, Y. Dong, and J. Tang. AgentBench: Evaluating LLMs as agents. In *ICLR*, 2024.
- [6] A. El hattami, M. Thakkar, N. Chapados, and C. Pal. WebArena Verified: Reliable evaluation for web agents. In *Workshop on Scaling Environments for Agents (NeurIPS)*, 2025.
- [7] S. Kara, F. Faisal, and S. Nath. WAREX: Web agent reliability evaluation on existing benchmarks. arXiv:2510.03285, 2025.
- [8] B. Zhao, D. Srikanth, Y. Wu, and Z. Jiang. SpecBench: Measuring reward hacking in long-horizon coding agents. arXiv:2605.21384, 2026.
- [9] J. Skalse, N. H. R. Howe, D. Krashennnikov, and D. Krueger. Defining and characterizing reward hacking. In *NeurIPS*, 2022.
- [10] V. Krakovna, J. Uesato, V. Mikulik, M. Rahtz, T. Everitt, R. Kumar, Z. Kenton, J. Leike, and S. Legg. Specification gaming: the flip side of AI ingenuity. DeepMind blog, 2020.
- [11] Y. Zheng, Y. Hu, T. Yu, and A. Quinn. AgentSight: System-level observability for AI agents using eBPF. In *Workshop on Practical Adoption Challenges of ML for Systems*, pp. 110–115, 2025. arXiv:2508.02736.
- [12] L. Dong, Q. Lu, and L. Zhu. AgentOps: Enabling observability of LLM agents. arXiv:2411.05285, 2024.
- [13] R. Koohestani. AgentGuard: Runtime verification of AI agents. In *IEEE/ACM International Conference on Automated Software Engineering Workshops (ASEW)*, pp. 74–77, 2025.
- [14] V. Bengs, E. Hüllermeier, and W. Waegeman. Pitfalls of epistemic uncertainty quantification through loss minimisation. In *Advances in Neural Information Processing Systems*, volume 35, pp. 29205–29216, 2022.
- [15] A. Der Kiureghian and O. Ditlevsen. Aleatory or epistemic? Does it matter? *Structural Safety*, 31(2):105–112, 2009.
- [16] E. Hüllermeier and W. Waegeman. Aleatoric and epistemic uncertainty in machine learning: an introduction to concepts and methods. *Machine Learning*, 110(3):457–506, 2021.
- [17] E. Hüllermeier, S. Destercke, and M. H. Shaker. Quantification of credal uncertainty in machine learning: a critical analysis and empirical comparison. In *Proceedings of the 38th Conference on Uncertainty in Artificial Intelligence (UAI)*, PMLR volume 180, pp. 548–557, 2022.
- [18] C. F. Manski. *Partial Identification of Probability Distributions*. Springer, 2003.
- [19] G. Shafer. *A Mathematical Theory of Evidence*. Princeton University Press, 1976.
- [20] P. Walley. *Statistical Reasoning with Imprecise Probabilities*. Chapman and Hall, London, 1991.
- [21] R. A. Howard. Information value theory. *IEEE Transactions on Systems Science and Cybernetics*, 2(1):22–26, 1966.
- [22] A. Agrawal, B. Amos, S. Barratt, S. Boyd, S. Diamond, and J. Z. Kolter. Differentiable convex optimization layers. In *NeurIPS*, 2019.
- [23] P. Z. Wang, J. Liang, S. Chen, F. Fioretto, and S. Zhu. Gen-DFL: Decision-focused generative learning for robust decision making. arXiv:2502.05468, 2025.
- [24] Z. Zhao, C. Yeh, L. Kong, and K. Wang. Diffusion-DFL: Decision-focused diffusion models for stochastic optimization. In *ICLR*, 2026.
- [25] J. Xie, H. Wu, Q. Chen, X. Su, and Y. Chen. Decision-weighted flow matching for contextual stochastic optimization. arXiv:2606.16790, 2026.
- [26] L. S. Shapley. A value for n -person games. In *Contributions to the Theory of Games II*, pp. 307–317. Princeton, 1953.
- [27] M. Sundararajan, K. Dhamdhere, and A. Agarwal. The Shapley–Taylor interaction index. In *ICML*, 2020.
- [28] T. Fissler and J. F. Ziegel. Higher order elicibility and Osband’s principle. *Annals of Statistics*, 44(4):1680–1707, 2016.
- [29] T. Gneiting. Making and evaluating point forecasts. *JASA*, 106(494):746–762, 2011.
- [30] B. Amos and J. Z. Kolter. OptNet: Differentiable optimization as a layer in neural networks. In *ICML*, 2017.
- [31] A. Ang and A. Timmermann. Regime changes and financial markets. *Annual Review of Financial Economics*, 4:313–337, 2012.
- [32] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster. Conformal risk control. arXiv:2208.02814, 2023.
- [33] N. Astorga, T. Liu, Y. Xiao, and M. van der Schaar. Autoformulation of mathematical optimization models using LLMs. In *ICML*, 2025.

- [34] D. B. Brown. Large deviations bounds for estimating conditional value-at-risk. *Operations Research Letters*, 35(6):722–730, 2007.
- [35] A. Butler and R. H. Kwon. Integrating prediction in mean-variance portfolio optimization. *Quantitative Finance*, 23(3):429–452, 2023.
- [36] P. F. Christoffersen. Evaluating interval forecasts. *International Economic Review*, 39(4):841–862, 1998.
- [37] E. Delage and Y. Ye. Distributionally robust optimization under moment uncertainty. *Operations Research*, 58(3):595–612, 2010.
- [38] A. Desai, C. Freeman, Z. Wang, and I. Beaver. TimeVAE: A variational autoencoder for multivariate time series generation. *arXiv:2111.08095*, 2021.
- [39] F. X. Diebold and R. S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.
- [40] P. Donti, B. Amos, and J. Z. Kolter. Task-based end-to-end model learning in stochastic optimization. In *NeurIPS*, 2017.
- [41] A. N. Elmachtoub and P. Grigas. Smart “predict, then optimize”. *Management Science*, 68(1):9–26, 2022.
- [42] I. Gibbs and E. Candès. Adaptive conformal inference under distribution shift. In *NeurIPS*, 2021.
- [43] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- [44] P. Grünwald, R. de Heide, and W. Koolen. Safe testing. *JRSS-B*, 2024.
- [45] J. D. Hamilton. A new approach to the economic analysis of nonstationary time series. *Econometrica*, 57(2):357–384, 1989.
- [46] S. Holt, M. Ruiz Luyten, A. Berthon, and M. van der Schaar. G-Sim: Generative simulations with large language models and gradient-free calibration. In *ICML*, 2025.
- [47] S. Holt, M. Ruiz Luyten, T. Pouplin, and M. van der Schaar. Fact-augmented lookahead planning for LLM agents. In *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR volume 300, 2026. [arXiv:2506.09171](https://arxiv.org/abs/2506.09171).
- [48] S. R. Howard, A. Ramdas, J. McAuliffe, and J. Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics*, 49(2):1055–1080, 2021.
- [49] C.-J. Kim and C. R. Nelson. Dynamic linear models with Markov-switching. *Journal of Econometrics*, 60(1–2):1–22, 1994.
- [50] M. Kollovich, A. F. Ansari, M. Bohlke-Schneider, J. Zschiegner, H. Wang, and Y. Wang. Predict, refine, synthesize: Self-guiding diffusion models for probabilistic time series forecasting (TSDiff). In *Advances in Neural Information Processing Systems*, 2023.
- [51] P. H. Kupiec. Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2):73–84, 1995.
- [52] M. López de Prado. Building diversified portfolios that outperform out of sample. *Journal of Portfolio Management*, 42(4):59–69, 2016.
- [53] M. López de Prado. *Advances in Financial Machine Learning*. Wiley, 2018.
- [54] O. Ledoit and M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.
- [55] B. Lim, S. Ö. Arik, N. Loeff, and T. Pfister. Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4):1748–1764, 2021.
- [56] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch. Multi-agent actor-critic for mixed cooperative-competitive environments. In *NeurIPS*, 2017.
- [57] H. Markowitz. Portfolio selection. *Journal of Finance*, 7(1):77–91, 1952.
- [58] A. Ramdas, P. Grünwald, V. Vovk, and G. Shafer. Game-theoretic statistics and safe anytime-valid inference. *Statistical Science*, 38(4):576–601, 2023.
- [59] K. Rasul, A.-S. Sheikh, I. Schuster, U. Bergmann, and R. Vollgraf. Autoregressive denoising diffusion models for multivariate probabilistic time series forecasting. In *ICML*, 2021.
- [60] R. T. Rockafellar and S. Uryasev. Optimization of conditional value-at-risk. *Journal of Risk*, 2(3):21–41, 2000.
- [61] D. Salinas, V. Flunkert, J. Gasthaus, and T. Januschowski. DeepAR: Probabilistic forecasting with autoregressive recurrent networks. *International Journal of Forecasting*, 36(3):1181–1191, 2020.
- [62] Y. Tashiro, Y. Song, J. Song, and S. Ermon. CSDI: Conditional score-based diffusion models for time series imputation. In *NeurIPS*, 2021.
- [63] P. S. Thomas and E. Learned-Miller. Concentration inequalities for conditional value at risk. In *ICML*, 2019.
- [64] V. Vovk, A. Gammerman, and G. Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- [65] Y. Wang and F. Gao. Deviation inequalities for an estimator of the conditional value-at-risk. *Operations Research Letters*, 38(3):236–239, 2010.
- [66] X. Wang et al. Executable code actions elicit better LLM agents. In *ICML*, 2024.
- [67] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. ReAct: Synergizing reasoning and acting in language models. In *ICLR*, 2023.
- [68] J. Yoon, D. Jarrett, and M. van der Schaar. Time-series generative adversarial networks. In *NeurIPS*, 2019.
- [69] M. Zaffran, O. Féron, Y. Goude, J. Josse, and A. Dieuleveut. Adaptive conformal predictions for time series. In *ICML*, 2022.

Appendix

The appendix is supplementary: it records specifications, full tables, derivations and boundary statements for the claims made in the main body.

A Verification-audit principles, in full

This case study motivates five verification-audit principles, which may transfer beyond it. Verify the opportunity before verifying the system: our L1 oracle was a representation-restricted evaluation protocol, not a weak baseline. Verify changes by intervention, since a null after an upstream update establishes nothing when downstream masking is possible. Treat interactions as first-class verification objects, because component-wise checks can obscure joint-repair patterns; in the present factorial the joint arm improves over baseline, while the interaction and conditional-contribution claims remain unresolved. Separate quality, admissibility and provenance, because a feasible or auditable action may still be poor. And verify the verifier: bypass, adaptive selection and invalid scoring rules are attack surfaces in their own right, and our own gate fails two of them.

B Claim–evidence map

Each question in §5 is answered by a specific, separately reported piece of evidence; Table 2 gives the resulting verdicts and their boundaries. Table 4 states, for each of the four questions posed in §4, the evidence used and the claim that evidence permits.

Table 4: **Verification questions posed of ARC-AGENT, the evidence used, and the claim that evidence permits.** Each row concerns a *different* construct. Sharpe is an operational proxy for risk-adjusted decision value, not a general measure of agent quality. Verdicts are collected in Table 2.

	Verification question	Evidence	Permitted claim
Q1	Was improvement attainable under an action set able to express it?	L0–L3 oracle Sharpe at fixed information, widening the action set (§5.1, Table 10); 24 seeds, paired, confirmatory	Measured headroom depends on the action representation, not on the information alone
Q2	Which component replacement recovers downstream value?	Stage-wise oracle replacement with the true regime given to every arm (§5.2, Table 5); 10–16 seeds, paired, confirmatory	A positive replacement identifies an observed bottleneck; a null one does not identify irrelevance
Q3	Does improved local fidelity verify downstream improvement?	Retention, directional cosine and vector error against paired Sharpe (§5.3, Table 7); 6 seeds, exploratory	Improved local scores do not establish downstream improvement
Q4	Do joint repairs reveal structure that one-at-a-time checks miss?	Complete 2^3 coalition table with Shapley and Shapley–Taylor summaries (§5.4, Tables 6, 15); 6 seeds, exploratory	The joint arm improves over the all-generator baseline; its conditional contribution and the interaction magnitude remain unresolved

Evaluation question	Evidence
Is meaningful improvement attainable?	L0–L3 oracle hierarchy (§5.1)
Did the full agent capture that improvement?	static risk frontier, paired contrasts (§5.1)
Which stage forfeited the improvement?	stage-wise oracle replacements (§5.2)
Can a module be masked by another?	complete 2^3 channel coalition table (§5.4)
Does the runtime boundary preserve admissibility?	per-action feasibility audits (§6)
Is the action traceable?	solver records and active-set logs (App. K)
Can the verifier itself fail silently?	A3 gate-bypass attack (§6)
Which claims remain unsupported?	unmet score identification (§4.4); unexecuted A5

C Oracle-replacement intervention outcomes

Table 5 reports the stage-wise interventions summarised in §5.2. Every row is given the true latent regime, so detection is never the confound, and rows are paired on common random numbers. “% of gap recovered” is measured against the L2 reference policy, not against an upper bound: L2 optimises mean–variance per regime, so a CVaR allocator handed the clean scenario law can exceed it, which is why I1 exceeds 100%. The remaining interventions of Fig. 2(b) (scenario count, shrinkage) are reported inline in §5.2.

Table 5: Oracle-replacement interventions (10 seeds, paired, common random numbers; true regime given to every row). Only a full generator bypass recovers the measured gap; no other intervention recovers a detectable share of it. The factorial of Table 6 subsequently separates that single bypass into the conditional-mean, covariance and scenario tail/dependence channels; the two experiments use different baselines and seed counts and their levels are not directly comparable (§5.4). “% of gap recovered” is $(\text{Sharpe} - \text{Sharpe}_{I0})/(\text{Sharpe}_{L2} - \text{Sharpe}_{I0})$. $\pm$ values are across-seed standard errors over 10 seeds under common random numbers. *L2 is a reference policy, not an upper bound*: it optimises mean–variance per regime, so a CVaR allocator given the clean scenario law can exceed it.

Intervention	Sharpe	Turn.%	% of gap recovered
I0 baseline (scenario moments, cap 0.20)	0.465 ± 0.090	19.7	n/a
I1 + clean regime-conditional scenarios and moments (<i>full generator bypass</i>)	1.141 ± 0.127	18.8	+121%
I2 + no turnover cap	0.377 ± 0.084	60.2	−16%
L2 regime-weight reference policy	1.023 ± 0.111	8.8	n/a

The allocator is not the bottleneck (detail for §5.2). We conjectured that the CVaR term, roughly $33\times$ the return term at a daily horizon, dominates the objective and leaves the solution insensitive to the conditional mean. We tested it, and it is false: swapping bull- for crisis-regime scenarios moves the QP’s solution by 118% of the oracle’s own rotation, aligned with it at $\cos(\Delta w_{QP}, \Delta w_{\text{oracle}}) = +0.75$. The optimiser would therefore have used a correct conditional mean had one reached it, which is what makes the generator, not the allocator, the observed bottleneck of §5.2.

D Factorial over decision functionals

The allocator reads the scenario law through three separable channels, so we can swap each independently between the generator’s output and clean regime-conditional history. Table 6 reports all eight cells of the 2^3 design, paired against the all-generator baseline. The two-channel rows are the informative ones: covariance adds nothing to any other repair, so the design collapses to a mean $\times$ scenarios interaction on a nuisance third factor. Note the asymmetry in the single-channel intervals: the scenario-only effect is $+0.079 \pm 0.066$, an interval that excludes zero, so that channel does deliver a modest standalone repair; the mean-only effect is $+0.020 \pm 0.206$, which is wide enough not to exclude the joint effect either. Descriptively, the joint-arm point estimate is larger than either single-arm point estimate. Because the paired contrast between the joint and scenario-only arms is unavailable from the retained aggregate summaries, this comparison does not establish a nonzero conditional mean contribution or superadditivity.

The contrast that bears most directly on the mean channel is $v(\{\text{mean}, \text{scenario}\}) - v(\{\text{scenario}\}) = +0.221 - 0.079 = +0.142$: what the clean conditional mean adds *once the scenario channel is already repaired*. We report the point estimate only. Its seed-bootstrap interval would be a paired contrast between two repaired cells rather than against the common baseline, and the per-seed pairs those two cells require were not retained, so the interval cannot be recovered from the available aggregate summaries. We therefore report this contrast descriptively and do not use it to support the interaction claim.

Reconciling Table 5 with Table 6. These are different experiments and their levels are not directly comparable, so we state the relationship explicitly. I1 is a *full generator bypass*: the allocator is handed the clean regime-conditional empirical scenario law (its scenario set together with the moments computed from it) in place of anything the diffusion model produces. The 2^3 factorial instead decomposes that single bypass into its three separable channels and swaps them independently. Consequently (i) I1 is the union of all three factorial repairs plus whatever the bypass changes that the three-channel parameterisation does not capture, and (ii) the two arms are run at different seed counts (10 vs. 6) against differently-constructed baselines (0.465 for I0, 0.635 for the all-generator factorial cell), which are themselves trained separately, as elsewhere in this paper (§5.4). The gap between I1’s 1.141 and the factorial’s all-three cell at 0.853 is therefore *not* evidence that the channels fail to compose; it is the combined effect of a wider intervention, a different baseline, and independent training runs. No paired arm is available to decompose that difference: with independently trained

Table 6: **The complete 2^3 factorial: all eight cells.** Paired ΔSharpe against the all-generator baseline (6 seeds, common random numbers, 95% CI over seeds). Reading down the single-channel rows, covariance is worth nothing; reading across the pairs, *adding* covariance to any other repair changes nothing (mean +0.020 versus mean+covariance +0.020; scenarios +0.079 versus scenarios+covariance +0.079), while mean+scenarios recovers essentially the whole effect and adding covariance on top of it does not improve matters (+0.221 vs. +0.218). At six seeds the design is **underpowered to distinguish superadditivity from additivity**. Descriptively the joint-arm estimate exceeds either single-arm estimate; because the paired joint-versus- scenario-only contrast cannot be recovered from the retained summaries, neither superadditivity nor a nonzero conditional mean contribution is established. The interaction estimate and its seed-bootstrap interval are reported in Table 15; at six seeds that interval includes zero. *These cells report Sharpe only.* We did not record MaxDD or turnover per cell, so the risk-axis check we insist on elsewhere (§5.1, §6) has not been applied to the factorial: a channel repair that moved a cell along the risk frontier rather than off it would not be visible here.

Channels replaced with clean regime-conditional history	Sharpe	ΔSharpe	95% CI
none (all generator; baseline)	0.635	n/a	n/a
covariance $\hat{\Sigma}$ only	0.634	−0.001	±0.001
conditional mean $\hat{\mu}$ only	0.655	+0.020	±0.206
scenarios only (tail + dependence)	0.713	+0.079	±0.066
mean + covariance	0.654	+0.020	±0.212
scenarios + covariance	0.714	+0.079	±0.066
mean + scenarios	0.856	+0.221	±0.160
all three	0.853	+0.218	±0.164

Table 7: **Retention metrics.** $\Delta\mu^{\text{true}}$ is the true bull-minus-crisis conditional mean vector and $\Delta\hat{\mu}$ the generator’s. The first row is the headline 29%; the middle rows are the same scalar computed in other arms; the last two rows are the *directional* and *vector* diagnostics that show why the scalar is insufficient.

Quantity	Definition	Reported values	Arm
Mean-gap retention	$\ \Delta\hat{\mu}\ _1 / \ \Delta\mu^{\text{true}}\ _1$	29% (90/313 bps)	deployed pipeline
same, step sweep	as above	19/30/76% ($S=20/50/100$)	step-count sweep
same, repair arm	as above	47% $\rightarrow$ 186%	repair arm
Directional similarity	$\cos(\Delta\hat{\mu}, \Delta\mu^{\text{true}})$	0.59 $\rightarrow$ 0.97	repair arm
Size ratio	$\ \Delta\hat{\mu}\ _2 / \ \Delta\mu^{\text{true}}\ _2$	208% (overshoot)	repair arm
Relative vector error	$\ \Delta\hat{\mu} - \Delta\mu^{\text{true}}\ _2 / \ \Delta\mu^{\text{true}}\ _2$	0.82 $\rightarrow$ 1.18	repair arm

baselines and different seed counts, the remaining discrepancy is not identifiable from the reported runs, and the boundary is recorded as item (ii) of App. L.

Mechanisms behind the mean collapse (detail for §5.3). The collapse implicates two candidate contributors to the observed retention loss. *Truncated reverse diffusion:* mean-gap retention is 19% at $S=20$ steps, 30% at our preregistered $S=50$ and 76% at $S=100$. A *starved conditioning class:* oversampling crisis $15\times$ lifts retention 30% $\rightarrow$ 53%. The reported η sweep does not support the tail-weighted loss as the primary driver of the retention loss: retention is flat in η and is 29% even at $\eta=0$. Repairing both mechanisms raises retention 47% $\rightarrow$ 186% within the repair arm and moves the paired Sharpe by $+0.075 \pm 0.174$. Arms are trained separately, so unrepaired retention differs slightly across them (29% deployed, 30% step-sweep, 47% repair); we always quote an arm against its own baseline.

The retention quantities, defined. Three distinct quantities appear above and are easy to conflate, so Table 7 defines each one. All are computed with the true regime given to the generator; none is a decision metric.

Table 8: **Does the complete system deliver genuine improvement at matched risk?** (**exploratory**, 5 seeds; $\pm$ is the across-seed standard deviation, not a paired contrast; those are in Tables 5–6.) The static frontier is the reference: “Excess” is Sharpe relative to the frontier at the same realised MaxDD, the comparator being the best static blend within the restricted EW–MinVar family. ARC-AGENT does not outperform the frontier, whereas the regime-conditional bootstrap (ten lines, no generative model) does, which is what shows the environment contains usable regime information. The –CVaR ablation lies beyond the frontier’s risk endpoint (off-supp.), as does mean–variance (Table 9).

Method	Sharpe	MaxDD%	Turn.%	Frontier	Excess
Equal weight	0.74 ± 0.46	18.5	0.0	0.74	+0.00
Minimum variance	0.37 ± 0.32	8.5	2.8	0.38	−0.01
Gaussian $\rightarrow$ QP	0.41 ± 0.27	12.1	19.7	0.60	−0.19
Block bootstrap $\rightarrow$ QP	0.44 ± 0.28	10.3	19.6	0.52	−0.08
Regime-cond. bootstrap $\rightarrow$ QP	0.85 ± 0.10	10.9	19.6	0.53	+0.32
ARC-AGENT (full)	0.56 ± 0.29	10.5	18.1	0.60	−0.04
– verifier	0.41 ± 0.29	9.7	18.3	0.48	−0.07
– CVaR term	0.84 ± 0.50	19.9	18.4	<i>off-supp.</i>	n/a

Table 9: The five remaining implemented baselines (**exploratory**, 5 seeds; mean $\pm$ across-seed standard deviation). None reaches the regime-conditional bootstrap’s 0.85. Mean–variance attains the highest Sharpe of the five but at 19.2% drawdown, beyond the static frontier’s risk endpoint, the same risk-axis confound the frontier test exists to expose, and the reason it is not evidence against our conclusion.

Method	Sharpe	MaxDD%	Turn.%
Risk parity	0.52 ± 0.30	11.4	1.5
Mean–variance	0.71 ± 0.50	19.2	12.0
Hierarchical risk parity	0.36 ± 0.29	8.7	3.2
Historical CVaR	0.45 ± 0.29	10.1	19.6
DRO-CVaR	0.34 ± 0.27	9.7	19.9

E End-to-end comparison at matched risk

This table supports the claim that end-to-end performance alone is insufficient. It is not a performance result in its own right, which is why it sits here (§5.5).

F Remaining baselines

Table 9 reports the five implemented baselines not shown in Table 8 or Table 11, so that all eleven are accounted for. They are run under the same box, turnover cap and costs as every other method, on the same paths. Like Table 8 these are **exploratory** five-seed results and are not paired contrasts.

G Oracle hierarchy

The hierarchy of §5.1 in full. All four oracles have perfect knowledge of the latent regime and run under identical box, turnover and cost settings; they differ only in the action set they are permitted to express that knowledge through. Figure 3 plots them against the static EW/MinVar frontier, so the headroom is read as a *vertical* gap at matched drawdown.

H Red-team results

Full numbers for the four *executed* attacks summarised at verification-surface level in Table 3 of §6. A5 (constraint injection against the validator) is registered and was not run, so it has no row here and no claim is made about it. The two comparison rows at the foot of the table are the rule-based weights-emitting controls of App. Q, standing in for an agent that emits weights directly with no

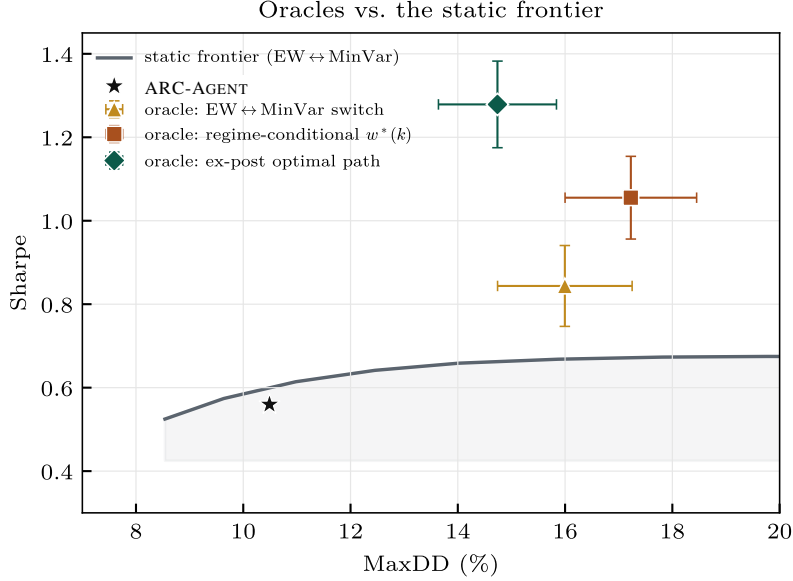


Figure 3: Oracles versus the static EW/MinVar frontier (24 seeds; bars are ± 1 s.e.; confirmatory, paired under common random numbers; Sharpe at $rf=1.6\%$). Construct: attainable decision value of perfect latent-state information under each action representation. The wider L2 and L3 action representations exhibit positive matched-risk oracle headroom in U3. ARC-AGENT lies close to the evaluated static frontier under the reported matched-risk comparison. Unresolved: whether this headroom transfers beyond U3 or beyond the four action sets shown.

Table 10: Oracle hierarchy (24 seeds, paired). **All Sharpe levels in this table are computed at the preregistered $rf=1.6\%$, not at the $rf=0$ used for the primary U3 metrics (App. Q); the paired L2–L0 contrast is $+0.28 \pm 0.08$ here and $+0.248$ at $rf=0$ (§5.1).** “Excess” is Sharpe over the static EW/MinVar frontier *at the same drawdown*, a different quantity from that paired contrast. The action set decides the answer: L1 is near-tautological, while L2, the oracle appropriate to a regime-conditional allocator, shows positive paired headroom under the wider action representation.

	Oracle over. . .	Sharpe	MaxDD%	Excess
L0	(no regime knowledge)	0.78	12.4	0.00
L1	two-point action set (EW↔MinVar)	0.84	16.0	+0.18
L2	regime-conditional weights $w^*(k)$	1.06	17.2	+0.38
L3	ex-post optimal weight path	1.28	14.7	+0.62
	ARC-AGENT	0.56	10.5	−0.04

solver to configure; they are *not* language-model agents, and no LLM was executed anywhere in this work.

I Why scalar local signals are insufficient (detail for §5.3)

Does any single upstream score verify downstream decision quality? Ours does not. Separate step-count and crisis-oversampling sweeps implicate two candidate contributors to the observed retention loss: truncated reverse diffusion and limited representation of the crisis conditioning class. The combined repair targets both contributors and therefore does not isolate their individual downstream effects; the reported η sweep does not support the tail-weighted loss as the primary driver (App. D). Repairing both raises retention **47% → 186%** while the paired Sharpe moves **$+0.075 \pm 0.174$** , with an interval spanning zero; the repaired generator (0.63) still loses to the bootstrap (0.85). The vector diagnostics show why: the repair fixes the *direction* of the regime mean-shift ($\cos 0.59 \rightarrow 0.97$) but *overshoots* its size (208%), so the relative error of the mean vector

Table 11: Red-team results, **measured** (5 seeds; $\pm$ is the across-seed standard deviation, not a paired contrast). Feasibility and provenance survive every *executed* attack (A1–A4), but so does feasibility for an agent that merely projects, which is why the surviving claim concerns auditability. **“Infeas.” is a post-hoc audit, not a construction guarantee:** every committed weight vector is checked against the budget (10^{-4}), box (10^{-6}) and turnover (10^{-4}) tolerances, and the column counts the rebalances that fail (App. O). **“Solver record”** reports whether the underlying convex solve produced a complete active-set and KKT record; it does *not* imply that the post-processed committed vector is itself KKT-optimal. **A5 (validator injection) was not executed and has no row.** The two comparison rows are rule-based weights-emitting controls; no LLM was executed.

	Attack	Sharpe	MaxDD%	Infeas.	Solver record
ARC-AGENT	none (base)	0.56 ± 0.29	10.5	0	100%
	A1 corrupted controller	0.50 ± 0.41	12.4	0	100%
	A2 poisoned memory	0.52 ± 0.28	10.4	0	100%
	A3 conformal-gate bypass	0.55 ± 0.29	9.9	0	100%
	A4 adversarial controller (injection proxy)	0.83 ± 0.38	16.7	0	100%
	weights-emitting agent (no solver)	0.50 ± 0.42	25.7	345	0%
	weights-emitting agent + projection	0.72 ± 0.40	19.4	0	0%

gets *worse* ($0.82 \rightarrow 1.18$). Distributional fidelity, conditional mean retention, verifier pass rate and downstream utility are not interchangeable, and a module can pass a local verification metric while failing the system-level objective. Our own measurements do not support *decision-functional retention* as the missing diagnostic either. This is the decision-focused-learning critique (41, 40) in sharp form, and it constrains what a remedy must target: not a better distributional score — likelihood, the verifier’s calibration tests, or even the conditional mean the optimiser literally reads — but the decision itself (App. M).

J Preregistered verdicts in full (detail for §5.5)

Table 8 gives the comparison for the arms that isolate scenario quality, and all five preregistered verdicts follow, each against its frozen failure condition. **H1 (tail control): inconclusive, not refuted.** ARC-AGENT’s MaxDD (10.5) versus a block bootstrap (10.3) is a paired +0.2pp against a design with little power at that scale; what we *can* say is that the diffusion model shows no detectable benefit even with perfect conditioning. **H1b (does the CVaR term do the work?): supported, with a twist.** Removing it inflates MaxDD $10.5 \rightarrow 19.9$ and raises Sharpe to 0.84, the highest in the table: a drawdown reduction bought with a real Sharpe cost, precisely the risk-axis trade the frontier test exposes. **H2 (union-bound correction): not exercised.** The correction never binds, so nothing here tests Prop. 2. **H3 (containment of corrupted intent): not supported (§6).** **H4: not executed.** No LLM controller was run.

K Active-set analysis from the logged duals

Which constraints actually bind (using the logged duals). Provenance is worth more than a 100% checkbox only if the KKT record is used. We therefore report active-set frequencies from the logged duals: in the illustrative run the turnover cap is slack in calm regimes (mean $\|\Delta w\|_1 \approx 8\%$ vs. the 20% cap) but *binds in* $\sim 38\%$ of *crisis-onset rebalances*, when the agent wants to de-risk faster than the cap allows; the box binds on the volatility proxy and short-duration sleeves during flights-to-quality. This is the one genuinely informative thing provenance yields, and it also sharpens the case for executing A5: the feasibility guarantee is not guarding an idle door. The turnover constraint is exactly what an attacker (or a panicked controller) would most want to relax, and it is the one the validator refuses to expose. A5 remains registered and unexecuted (§6).

What the record does and does not give. The KKT record certifies feasibility and stationarity of the *solver output* w^{sol} for the problem it was solved against. The subsequently committed action w_t , obtained from w^{sol} by clipping and renormalisation (App. O), *passed the specified post-hoc feasibility audit*, but we do not claim it retains the solver’s KKT optimality certificate (Prop. 3). Neither claim, by itself, re-derives w_t from scratch. *Deterministic replay* additionally requires the realised scenario

matrix (or the seed and generator state that produced it), the model checkpoint hash, the solver name, version and tolerances, and the tie-breaking rule used when the active set is degenerate. Our log records these alongside the certificate, but the guarantee we state and prove is the verification claim above; replay is an implementation property of the release, not a theorem.

L Boundaries of established knowledge

This appendix is organised by *claim* rather than by experiments that “should” be run. Each entry states the exact limit of the existing evidence, and none is a promise to fix it for this paper. With the claim–evidence ledger (Table 2), these entries record what the audit did not establish.

1. Domain boundary. Every empirical statement is scoped to U3 and to one decision architecture. We do not claim “generative modelling is a bottleneck in modular decision systems” in general; we claim that *this* conditional diffusion model, trained this way, forfeits the conditional mean on *this* benchmark, that the repair arm targets two plausible contributors to the collapse, and that repairing them does not repair the decision. The regime-conditional bootstrap’s advantage is partly favoured by U3’s stationary within-regime conditionals: a resampler inherits the conditional law by construction, so it is close to a near-oracle generator here. Establishing where conditional *generation* could pay requires sweeping within-regime nonstationarity, regime overlap and duration heterogeneity, which we have not done.

2. Masking boundary. Perception cannot be judged independently while the generator ignores its conditioning. Perfect detection does not help *through the current generator*; that does not establish perception is unimportant, and perception may matter once the generator is fixed. The factorial detector $\times$ generator matrix that would separate the two was not run, so the verdict in Table 2 is *unresolved*: neither supported nor refuted.

3. Intervention boundary. The stage interventions (Table 5) and the channel factorial (Table 6) use different baselines and seed counts and were trained separately, so the difference between the full generator bypass and the all-three-channel repair *cannot be decomposed* (§5.4, App. D). Rerunning both against one baseline under common random numbers at a single seed count would identify whether the three channels exhaust the bypass.

4. Risk-axis boundary. Table 6 reports Δ Sharpe alone: the factorial lacks MaxDD and turnover per cell, so we cannot show that a channel repair moves a portfolio *off* the risk frontier rather than along it, the confound our own frontier test exists to expose and which we apply everywhere else. The risk-axis position of each factorial cell is therefore not identified by the reported runs.

5. Architecture boundary. One conditional generator family is evaluated. Without conditional VAE, normalising-flow, plain conditional diffusion and moment-regularised variants, the localisation result remains single-architecture.

6. Auditability boundary. The surviving governance claim is an *argument from the available records*, not a measurement: we have not scored whether an auditor can recover the binding constraint and the reason-for-trade from the log alone. This is the paper’s weakest link, since it is the one claim asserted without evidence.

7. Controller boundary. No LLM controller was executed. All measured runs use a **rule-plus-contextual-bandit controller**; the system under test is a planning loop over typed tool configurations, and A4 is an adversarial-*controller* proxy, not a live prompt injection. Nothing here bears on language-model reasoning or prompt-injection robustness.

8. Statistical boundary. Seed counts are 5–24 and inconsistent across tables. The factorial is the binding case: at six paired seeds the mean-only interval is very wide and the interaction interval includes zero, so the *magnitude* of the mean–scenario interaction is unresolved and the headline 55% is a point estimate. Twenty to thirty paired seeds under common random numbers would settle it, and App. R records that compute is not what is stopping us.

9. Certificate boundary. The implemented one-step score does not identify CVaR_α (§4.4, App. W), so the executed gate is an operational filter and not a validated decision-quality certificate. H2 was never exercised, so nothing in this paper tests the refinement correction either.

10. Threat-model boundary. Trusted-path compromise was not executed. A5 targets the validator, and the solver, post-solve cleanup and audit code are equally inside the trusted path; robustness to compromise of any of them is unevaluated, so the feasibility result of Table 3 is evidence about attacks that leave those components alone.

Future work. Decision-focused training of the scenario generator; a disaggregated retention analysis over the mean, covariance and scenario tail/dependence channels; a second decision domain (inventory, energy dispatch) to test whether the localisation protocol transfers; and conformal *risk control* on a monotone loss in place of coverage of a CVaR difference.

M Extended related work

Agentic systems and tool use. ReAct-style interleaving of reasoning and acting (67) and code-as-action agents (66) motivate our typed configuration interface. Recent work couples language models with formal optimisation: autoformulation of optimisation models (33) and generative simulation with gradient-free calibration (46) show that LLMs are most reliable when they *configure* a solver rather than replace it. Lightweight online memory without finetuning (47) informs our regime-indexed episodic store. We adopt this “LLM plans, solver acts, verifier gates” pattern as a governance principle rather than a convenience. A systems literature instruments running systems at stable boundaries rather than inside volatile application code (11, 12, 13); our validator, feasibility audit and KKT records are instances of that discipline, and the distinction we insist on is that *observability provides evidence; verification specifies what property that evidence supports*.

Regimes and structured generative time series. HMMs capture structural breaks and state dependence (45, 49, 31); we reuse their posteriors as conditioning signals. GAN/VAE and diffusion generators improve realism (68, 38, 59, 62), but are typically unconditioned and loosely coupled to decisions; ARC-AGENT instead couples a regime-conditioned, tail-weighted diffusion generator to the allocator as a callable, verifiable tool.

Decision-focused learning. A parallel literature learns predictive models *through* the optimiser rather than upstream of it: smart “predict-then-optimize” (41), task-based end-to-end learning (40), and differentiable optimisation layers (30, 22), with portfolio applications (35) and, more recently, decision-focused *generative* models (23, 24, 25), which target exactly the generator-to-decision coupling that our §5.3 measurement finds broken. ARC-AGENT is deliberately *not* end-to-end: it keeps the solver a fixed, certified, non-differentiated component so that its guarantees survive a compromised upstream, and treats decision-aware training as future work. The trade-off this involves is easy to overstate. It is *not* optimality versus auditability: an upstream model can be trained through a differentiable convex layer (30, 22) and still be deployed against the same fixed solver, which still emits the same KKT record, so auditability survives decision-focused training. What decision-focused training actually costs is *modularity*: the generator becomes coupled to a particular downstream program and to its training pipeline, and can no longer be swapped, certified, or attacked in isolation in the way §4.5 relies on. That coupling is the central design choice we are making, and this literature is the alternative branch. Our own measurement (§5.3) is an argument for paying that cost.

Distribution-free uncertainty quantification. The verifier is implemented as a conformal-style *operational evidence filter*. Conformal prediction (64) and conformal risk control (32) provide finite-sample, distribution-free coverage guarantees and are the right foundation for a valid act-time certificate, but the implemented score does not satisfy the identification conditions required for that interpretation (§4.4).

Robust allocation and governance. Distributionally robust and CVaR formulations address fat tails (60, 37) but hinge on scenario quality; we feed a regime-conditioned scenario set through an operational evidence filter into a convex allocator with Ledoit–Wolf shrinkage (54). Multi-agent coordination (56) inspires our role-based tools, but our emphasis is on *auditability*: the allocator’s

KKT system provides solver-side audit trails that align with the reliability/governance goals of structured-data agents.

N The U3 benchmark

U3 is a controllable synthetic regime-shift generator over d correlated instruments. A latent Markov chain with $K=3$ states (*bull*, *stagnation*, *crisis*) drives a state-dependent return law: per-state mean and covariance, with crisis drawn from a fat-tailed, left-skewed law whose lower-tail dependence exceeds its average correlation, so diversification degrades exactly when it is needed (Fig. 4b,c). There are $d=10$ instruments, three of them defensive. Returns follow a regime-dependent factor model driven by a skew- t market factor with shared mixing, so no regime is Gaussian: degrees of freedom fall from 12 (bull) through 7 (stagnation) to 3.5 (crisis) and skew steepens from 0 to -0.80 (App. O). Crisis is rare and persistent: the stationary crisis share is 4.79% and the mean crisis episode is about 45 trading days, longer than the 21-day rebalance interval; the evaluated 1,500-day development sequence realises 66 crisis days (4.4%). Four knobs are exposed: *crisis frequency* (the chain’s entry probability), *severity* (the crisis mean and tail index), *duration* (its self-transition probability), and *detectability* (*regime_separation*, which interpolates the per-state means toward a common value). At *regime_separation*=0 the states are indistinguishable from returns alone and HMM balanced accuracy equals chance, 1/3, by construction; this is the calibration check in Fig. 4(d). Because the latent state is recorded, regime inference can be scored directly and the oracle hierarchy of §5.1 can be computed exactly, neither of which is possible on historical data.

Horizon and splits. Each seed draws a single 2,200-day path. The first 756 days (≈ 3 years) are a **warm-up** used only to initialise the HMM and to fit the first generator; no trade is placed inside it. The remaining 1,444 days are evaluated walk-forward with a rebalance every 21 trading days, giving 69 **evaluated rebalances per seed**. Estimation is strictly causal: at each rebalance the generator and detector see only data up to that date, and the generator is refit every 1,000 trading days (2 refits per run), the cadence behind every reported number (Table 12, App. O). The generator, its default parameter values, and the seed registry are specified in Table 12, Table 14 and App. O; code and configuration files will be released upon acceptance.

Reproducibility summary. Table 12 collects every setting needed to reproduce the measured results in one place; App. O gives the full specification.

O Full experimental specification

Regime chain. The latent chain over {*bull*, *stagnation*, *crisis*} is built from per-regime persistences (0.992, 0.980, 0.978), with the exit mass split 0.93/0.07 from bull into stagnation/crisis, 0.88/0.12 from stagnation into bull/crisis, and 0.35/0.65 from crisis into bull/stagnation. This yields

$$P = \begin{pmatrix} 0.99200 & 0.00744 & 0.00056 \\ 0.01760 & 0.98000 & 0.00240 \\ 0.00770 & 0.01430 & 0.97800 \end{pmatrix}, \quad (2)$$

whose stationary distribution is (0.669, 0.283, 0.048). Crisis self-transition 0.978 gives a mean crisis episode of about 45 trading days, deliberately longer than the 21-day rebalance interval, since an episode shorter than the decision interval could not be rotated into by a monthly agent and the benchmark could not test the paper’s claim. The stationary crisis share of 4.79% implies 71.9 crisis days in a 1,500-day window; the evaluated path realises 66 (4.4%), an ordinary finite-path deviation. Realised and stationary prevalence need not coincide, and it is the realised figure that determines how much crisis data the generator actually saw.

Return law. Returns follow a regime-dependent factor model, $r_i = \mu_i(k) + \beta_i(k)F(k) + \epsilon_i$, where $F(k)$ is a skew- t market factor with a *shared* chi-square mixing variable, which is what produces lower-tail *dependence* rather than merely high correlation: assets co-crash. The factor has annualised drift (0.20, 0.01, -0.55), daily volatility (0.0060, 0.0095, 0.0210), degrees of freedom (12, 7, 3.5) and skew (0, -0.25 , -0.80) across bull, stagnation and crisis; residual cross-correlation among risky names is (0.10, 0.15, 0.30) and idiosyncratic volatility is 0.0035 daily. No regime is Gaussian: the

Table 12: **Reproducibility summary** for the measured (U3) experiments. All values are taken from the frozen preregistration and the implementation; the data-generating process, architecture and verifier are given in full in App. O.

Setting	Value
<i>Benchmark</i>	
Regimes K	3 (bull, stagnation, crisis); $K \in \{2, 3, 4\}$ searched
Instruments d	10, of which 3 defensive (negative crisis beta)
Transition matrix	Eq. (2); stationary (0.669, 0.283, 0.048)
Regime persistence	(0.992, 0.980, 0.978); mean crisis episode ≈ 45 days
Factor drift (annualised)	(0.20, 0.01, -0.55)
Factor daily volatility	(0.0060, 0.0095, 0.0210)
Skew- t degrees of freedom	(12, 7, 3.5); skew (0, -0.25 , -0.80)
Residual correlation	(0.10, 0.15, 0.30); idiosyncratic vol 0.0035
Crisis prevalence	4.79% stationary (71.9 crisis days per 1,500)
Simulated horizon	2,200 days per seed: 756-day warm-up, 1,444 evaluated
Rebalances	69 (monthly, every 21 trading days); embargo 21 days
<i>Generator and denoiser</i>	
Formulation	variance-preserving diffusion; predicts ε
Denoiser	two experts, each $45 \rightarrow 128 \rightarrow 128 \rightarrow 10$ with SiLU
Gate	$g_t = \sigma(\text{MLP}(\pi_t))$, $3 \rightarrow 32 \rightarrow 1$ with SiLU
Input dimension	$45 = d(10) + \text{time embedding}(32) + K(3)$
Parameter counts	23,690 per expert + 161 gate = 47,541 total
Diffusion steps S	50; swept over $\{20, 50, 100\}$
Noise schedule	linear β from 10^{-4} to 0.06 over S steps
Scenarios N	$\{256, 512\}$ searched, 512 default; 4,096 in the sampling intervention
Return winsorisation	$ r \leq 0.10$ (bounds M in the concentration bound)
<i>Training</i>	
Schedule	12 epochs per refit; batch 128; Adam; lr 2×10^{-3}
Refit cadence	every 1,000 trading days (2 refits per run)
Selection	deflated Sharpe, computed within the walk-forward
<i>Verifier</i>	
Method	adaptive conformal inference (42, 69)
Target miscoverage	$\delta = 0.10$ (90% long-run act-time coverage)
Budget	$B = 3$; per-attempt level $\delta/B \approx 0.033$
Score s_t	Definition 1; $ \text{realised tail loss} - \text{scenario CVaR}_\alpha $
Calibration window	60 scores (rolling); ACI step size 0.01
Fallback	stationary block bootstrap into the same CVaR-QP
<i>Decision and evaluation</i>	
Constraints	long-only, fully invested; box $[0, 0.25]$; turnover ≤ 0.20 per rebalance
Costs	8 bps proportional
CVaR level α	0.95 default; $\alpha \in \{0.90, 0.95, 0.99\}$ searched
Risk-free rate	$rf=0$ primary on U3; oracle hierarchy at $rf=1.6\%$ (App. Q)
Solver	CLARABEL via CVXPY (≥ 1.4), SCS fallback; no overrides set
Solver tolerances	CLARABEL gap/feas. tol. 10^{-8} , max_iter= 200; SCS $\varepsilon_{\text{abs}}=\varepsilon_{\text{rel}}=10^{-4}$, max_iters= 10^5
Hyperparameter grid	2,916 configurations $\times$ 5 seeds = 14,580 (App. T)
Ablation arms	–verifier, –memory, –MoE, –CVaR term, $B=1$
Seed registry	Table 14

tails thicken and the skew steepens monotonically from bull to crisis. Of the $d=10$ instruments, 3 are defensive, with near-zero normal beta and *negative* crisis beta (a flight-to-quality corner), and a lower unconditional carry (0.8% annual) than the risky names. Both properties are load-bearing: without a defensive corner the allocator is always fully invested in statistically identical assets, equal weight is optimal by construction, and regime-conditioning has nothing to earn; without the carry penalty, holding the defensive assets always would dominate and regime-timing would again be worthless.

Denoiser. Each of the two experts is a three-layer MLP $45 \rightarrow 128 \rightarrow 128 \rightarrow 10$ with SiLU activations, where the 45-dimensional input concatenates the noisy return vector ($d=10$), a 32-dimensional sinusoidal time embedding, and the $K=3$ regime posterior. The gate is $3 \rightarrow 32 \rightarrow 1$ with a sigmoid output, combining the experts as $\hat{\varepsilon}_\theta = (1 - g_t)\hat{\varepsilon}_\theta^{\text{base}} + g_t\hat{\varepsilon}_\theta^{\text{crisis}}$ so the weights sum to one by construction. Exact parameter counts are 23,690 per expert and 161 for the gate, 47,541 in total; the $-$ MoE ablation drops the crisis expert, leaving 23,851. Diffusion uses $S=50$ steps with β linear from 10^{-4} to 0.06. For the reported baseline and ablation table the generator is trained for 12 epochs per refit, batch 128, Adam at $\text{lr } 2 \times 10^{-3}$, and refit every 1,000 trading days, giving **two refits per run**. *Two other scripts in the released code, the smoke test and an early ablation sweep, use a different generator budget (25 epochs, refit every 504 trading days) and are not the source of any reported number; the values here are those behind Table 8, and they are the values used by every measured arm (diff_epochs=12, retrain_every=1000).*

Verifier, operationally. The score of Definition 1 is computed exactly as Definition 1 states. The verifier keeps a rolling window of the 60 most recent scores, takes the empirical $(1 - \beta_t)$ -quantile of that window as the certificate width q_t , targets miscoverage $\delta=0.10$, and updates the running level online by $\beta_{t+1} = \beta_t + \gamma_{\text{ACI}}(\delta - \mathbf{1}\{s_t > q_t\})$ with step size $\gamma_{\text{ACI}} = 0.01$ and $\beta_0 = \delta/B$. Attempts are capped at $B=3$ and each is evaluated at $\delta/B \approx 0.033$; the first passing attempt is accepted. If no attempt passes, or if the solve fails, the system falls back to a stationary block bootstrap whose scenarios are solved by the *same* CVaR-QP.

Why (C5) is not a technicality, and what would discharge it. The gap in (C5) is structural rather than a missing lemma. Our score is a single-realisation residual $|\ell_{t+1} - \widehat{\text{CVaR}}_\alpha|$, and $\widehat{\text{CVaR}}_\alpha$ is **not elicitable**: no scoring function has $\widehat{\text{CVaR}}_\alpha$ as its unique minimiser (29), so no one-sample residual can identify it, and no amount of conformal calibration on such a score can be made to bound $|\widehat{\text{CVaR}}_\alpha^{p_\theta} - \widehat{\text{CVaR}}_\alpha^{\mathbb{P}}|$. This is why we mark (C5) unproved rather than deferred: as specified, it is not provable.

There is, however, a constructive route, and we state it because it prescribes a concrete change to the verifier; no reported run is reinterpreted by it. The *pair* $(\text{VaR}_\alpha, \widehat{\text{CVaR}}_\alpha)$ is jointly elicitable, via the Fissler–Ziegel class of strictly consistent scoring functions (28). A verifier that scored the model’s $(\text{VaR}, \widehat{\text{CVaR}})$ pair under an FZ-consistent loss would inherit strict consistency, and under the accompanying strong-identifiability condition an excess-score bound converts into a bound on the population discrepancy, supplying exactly the bridge (C5) assumes. We did not run this variant, so we register it as the repair our own diagnosis implies, in the same spirit as §4.3: the protocol convicts the paper’s theory as well as its generator.

What s_t is and is not. s_t is an *observable* absolute one-step discrepancy between a scenario CVaR estimate and a single realised loss. It is not an estimate of the population quantity $|\widehat{\text{CVaR}}_\alpha^{p_\theta} - \widehat{\text{CVaR}}_\alpha^{\mathbb{P}}|$, and a single realised return cannot reveal that quantity. The link between the calibrated quantiles of $\{s_\tau\}$ and the population discrepancy is exactly the identification condition that the conditional certificate statement 1 assumes and that we do not prove (App. W).

Allocator and solver. The convex program (5) is solved through CVXPY with CLARABEL as the primary interior-point solver and SCS as the fallback, invoked only if the primary solve raises. No tolerance, iteration or polish setting is overridden anywhere in the code, so each solver’s compiled-in defaults apply. We state them here so the PDF is self-contained: CLARABEL uses absolute and relative duality-gap tolerances and a feasibility tolerance of 10^{-8} with a 200-iteration limit; SCS uses $\varepsilon_{\text{abs}} = \varepsilon_{\text{rel}} = 10^{-4}$ with a 10^5 -iteration limit. The released dependency specification pins CVXPY ≥ 1.4 , the first release in which CLARABEL is the default conic solver, and CLARABEL and SCS enter as transitive dependencies of that pin. The solver reports one of three outcomes per rebalance: a normal solve, a fallback solve, or, if no weight vector is returned, status FAILED, in which case the

committed action is the previous feasible portfolio w_{t-1} . FAILED did not occur in any reported run. Returns are winsorised at $|r| \leq 0.10$, which also bounds M in the concentration term of Lemma 2.

Post-solve cleanup: what it does and does not guarantee. The solver’s solution w^{sol} is clipped to the box and renormalised to sum to one before it is committed. This is numerical cleanup and a *no-op at an exact solution*, but it is post-processing, and we state its guarantee precisely because an earlier draft overstated it. Clipping alone gives $\ell \leq w_i \leq u$. The subsequent division by $\sum_i w_i$ restores the budget exactly and, with $\ell=0$ as here, preserves non-negativity, but it is **not** feasibility-preserving in general. If the clipped vector sums to $s < 1$ every coordinate is scaled up by $1/s$, which can lift one above u ; for $\ell > 0$ and $s > 1$ the symmetric violation is possible at the lower bound; and rescaling changes $\|w_t - w_{t-1}\|_1$, so the turnover cap can be breached too. Nor do we claim a general perturbation bound. Renormalisation rescales *every* coordinate by $1/s$, so the displacement it induces grows with the dimension and with how close the active coordinates sit to their bounds; bounding it would need assumptions we have not stated. What we rely on is that the solver residuals are numerically small, so the post-processing perturbation was small in the reported runs, and, decisively, on the explicit audit. No theoretical guarantee is doing any work here. The paper’s feasibility claim is therefore checked, not constructed: the “Infeas.” column of Table 11 is a post-hoc audit of every committed weight vector against the budget (10^{-4}), box (10^{-6}) and turnover (10^{-4}) tolerances, and it is zero for every executed arm and seed. That is evidence, not a guarantee. “Verifiable provenance” likewise means the solve is reconstructible from the log, not that a certificate has been re-derived for a post-processed vector: the logged KKT duals certify w^{sol} , not the clipped iterate.

The variant that would restore the guarantee (registered, not run). The clean design does not post-process at all. The trusted validator should check the budget, box and turnover residuals of w^{sol} against fixed tolerances and commit w^{sol} only if all three pass, invoking the predefined block-bootstrap fallback otherwise. That variant is better on two counts: every committed action is feasible to a stated tolerance by a check the *validator* owns rather than by a numerical convenience, and the committed vector is the solver’s own iterate, so the logged KKT record certifies the action actually taken and the w^{sol} -versus- w_t distinction of Prop. 3 disappears. A Euclidean projection onto the full constraint set would also restore feasibility, but it is a second optimisation whose output carries no KKT record from the first, so it reintroduces the gap it closes. We ran neither: every number in this paper comes from the clip-and-renormalise pipeline described above, and we register the change rather than retrofit it, exactly as we do for the (C5) scoring change above.

Action-space partition (detail for §4.5). The *untrusted*, controller-settable parameters are the objective and generator parameters $(K, N, \eta, q, \lambda, \gamma, \lambda_\mu, \alpha)$; none of them can enlarge the feasible set, so a corrupted controller can degrade decision *quality* but not admissibility. The *trusted*, validator-owned parameters are the constraint parameters (ℓ, u, τ) and the refinement budget B . The validator clips any untrusted value to its preregistered range and rejects any attempt to name a constraint parameter.

Controller. The controller is a **transparent rule plus a contextual bandit** (LinUCB-lite) over the preregistered grid; no LLM is involved. The rule keys off the crisis posterior $\pi_{t,\text{crisis}}$: above 0.35 it sets $\eta=3.0$, $\alpha=0.99$, $\gamma=1.5$, $q=0.10$ and $\lambda_\mu=4$ (risk-off); between 0.15 and 0.35 it sets $\eta=2.0$, $\alpha=0.95$, $\lambda_\mu=12$; below 0.15 it sets $\lambda_\mu=24$ (risk-on). The bandit then refines $\{\eta, \gamma, \lambda, \lambda_\mu\}$ within the grid, keeping per-(regime, knob, value) reward means and exploring uniformly with probability $\varepsilon = 0.15$. Every value it can emit is inside the preregistered ranges, and the validator clips anything outside them.

Regime detector. A Gaussian HMM (hmmlearn) with K states on z -scored features, `covariance_type="diag"`, `n_iter= 200`, `tol= 10-4`, and **4 random restarts** scored by log-likelihood with the best retained; restart seeds are `seed + 13r`. The offline regime-scoring used for the detectability calibration of Fig. 4(d) instead uses `covariance_type="full"` with `tol= 10-3` and the same restart count.

Baseline tuning. The scenario-fed baselines (historical CVaR, DRO-CVaR, Gaussian, stationary block bootstrap) are *untuned by construction*: they feed the same CVaR-QP at the same α , box, turnover cap and costs, so only scenario quality differs. The classical baselines take their standard

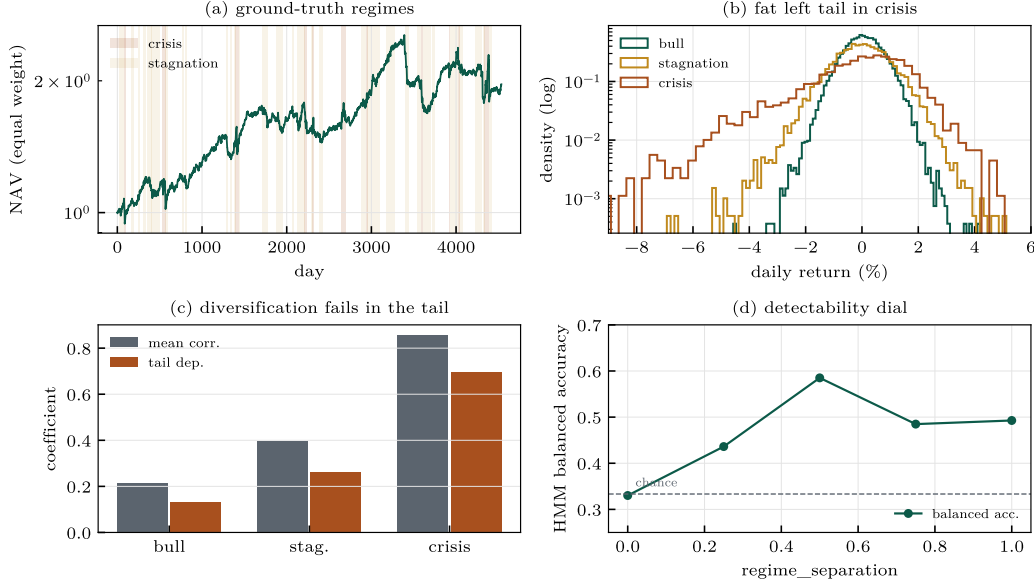


Figure 4: The U3 benchmark, drawn on an **illustrative 4,536-day path** from the same generator (panel (d) uses 2,600-day paths per seed); this is for visualising the data-generating process and is *not* the evaluated horizon, which is 2,200 days per seed (756 warm-up, 1,444 evaluated). **(a)** ground-truth regimes; **(b)** crisis returns are fat-tailed and left-skewed; **(c)** correlations *and* lower-tail dependence both spike in crisis, so diversification fails exactly when it is needed; **(d)** the detectability dial: at `regime_separation=0` balanced accuracy equals chance (1/3) by construction.

forms with no free parameters beyond the shared constraints, except HRP, which uses the standard single-linkage correlation-distance clustering, and the block bootstrap, whose block length is 21 days (one rebalance interval).

Attacks, operationally. A1 randomises λ_μ and γ within $[0, 30]$; A2 flips the HMM regime posterior; A3 bypasses the conformal gate (it does not touch the validator); A4 maximises turnover. None of the four modifies the trusted validator, which is why the feasibility result they support excludes validator compromise (A5, not executed).

P Problem setup, tools, and the agent loop

At rebalance date t , let $R_t \in \mathbb{R}^d$ be observed returns and $w_t \in \mathbb{R}^d$ portfolio weights satisfying the budget, box, and turnover constraints

$$\mathbf{1}^\top w_t = 1, \quad \ell \leq w_t \leq u, \quad \|w_t - w_{t-1}\|_1 \leq \tau. \quad (3)$$

Regimes follow a K -state Gaussian HMM with latent state S_t and posterior $\pi_{t,k} = \mathbb{P}(S_t=k \mid R_{1:t})$. All estimation is strict walk-forward. ARC-AGENT orchestrates three specialist tools, treated as fixed, trusted backbones; the controller and evidence-filter layer (§3) sits around them.

Tool A: Regime detector. Fit/update the HMM on $\{R_s\}_{s \leq t}$ and emit π_t and a context vector z_t encoding $\arg \max_k \pi_{t,k}$ and recent statistics.

Tool B: Scenario generator. A variance-preserving, regime-conditioned diffusion model with a tail-weighted objective. With $\tilde{\ell} = -\min_j r_j$ a portfolio-free adverse proxy,

$$\mathcal{L}_{\text{tail}} = \mathbb{E} \left[(1 + \eta \mathbf{1}\{\tilde{\ell} \geq Q_{1-q}(\tilde{\ell})\}) \|\varepsilon - \varepsilon_\theta(\cdot)\|_2^2 \right], \quad q \in [0.05, 0.10], \quad \eta \in [1, 3], \quad (4)$$

The denoiser is a two-expert (mixture-of-experts) network, $\hat{\varepsilon}_\theta = (1 - g_t)\hat{\varepsilon}_\theta^{\text{base}} + g_t\hat{\varepsilon}_\theta^{\text{crisis}}$ whose gate $g_t = \sigma(\text{MLP}(z_t))$ increases with the crisis posterior. At deployment it draws N scenarios $\{r_{t+1}^{(i)}\}$ conditioned on z_t .

Algorithm 1 ARC-AGENT: one rebalance at date t

```
1: input: history  $R_{1:t}$ , previous weights  $w_{t-1}$ , refinement budget  $B$ 
2:  $\pi_t, z_t \leftarrow \text{REGIMEDETECTOR}(R_{1:t})$  ▷ Tool A; strict walk-forward
3:  $a_t \leftarrow \text{CONTROLLER}(z_t, \text{memory})$  ▷ config only; validator clips to preregistered ranges
4:  $b \leftarrow 0$ 
5: repeat
6:    $\{r_{t+1}^{(i)}\}_{i=1}^N \leftarrow \text{GENERATOR}(z_t; a_t)$  ▷ Tool B
7:    $\text{pass} \leftarrow \text{VERIFIER}(\{r_{t+1}^{(i)}\}, \{s_\tau\}_{\tau < t}, \delta/B)$  ▷ conformal gate, Prop. 2
8:    $b \leftarrow b + 1$ ; if  $\neg \text{pass}$  then  $a_t \leftarrow \text{REFINE}(a_t)$ 
9: until  $\text{pass}$  or  $b = B$ 
10: if  $\neg \text{pass}$  then
11:    $\{r_{t+1}^{(i)}\} \leftarrow \text{STATIONARYBLOCKBOOTSTRAP}(R_{1:t})$  ▷ fallback
12: end if
13:  $(\hat{\mu}_t, \hat{\Sigma}_t) \leftarrow \text{blended, shrunk moments of } \{r_{t+1}^{(i)}\}$ 
14:  $w_t \leftarrow \arg \min \text{ of the convex QP (5) subject to (3)}$  ▷ Tool C
15:  $s_t \leftarrow \text{realized calibration score once } r_{t+1} \text{ is observed}$ 
16: log  $(\pi_t, a_t, \text{verifier outcome, active set, KKT duals})$ 
17: return  $w_t$  ▷ feasible by Prop. 3; KKT record is for  $w^{\text{sol}}$ 
```

Note on the indicator (corrected). An earlier draft printed Eq. (4) with $\mathbf{1}\{\tilde{\ell} \leq Q_q(\tilde{\ell})\}$, which up-weights the *least* adverse samples, the opposite of the intent. The *implementation* was always correct; the equation was not. We verified this rather than asserting it: the up-weighted set has mean return -107 bps and is $2.9\times$ enriched in crisis days, whereas the erroneous indicator would have selected samples averaging $+106$ bps. The audit script will be released upon acceptance. The 29% mean-collapse result is therefore not an artefact of an inverted indicator, and η is ruled out independently (§5.3).

Tool C: CVaR allocator. With per-scenario loss $\ell_i = -w^\top r_{t+1}^{(i)}$ and blended, shrunk moments $(\hat{\mu}_t, \hat{\Sigma}_t)$, solve the convex epigraph QP

$$\begin{aligned} \min_{w, \zeta, \{s_i\}} \quad & -\lambda_\mu \hat{\mu}_t^\top w + \gamma w^\top \hat{\Sigma}_t w + \zeta + \frac{1}{(1-\alpha)N} \sum_{i=1}^N s_i \\ \text{s.t.} \quad & s_i \geq 0, \quad s_i \geq \ell_i - \zeta, \quad i = 1, \dots, N, \\ & \mathbf{1}^\top w = 1, \quad \ell \leq w \leq u, \quad \|w - w_{t-1}\|_1 \leq \tau. \end{aligned} \tag{5}$$

The objective is a convex QP (quadratic $w^\top \hat{\Sigma}_t w$ plus a CVaR epigraph with $O(N)$ affine constraints); interior-point solves scale as $O(d^3 + Nd^2)$. We use the Rockafellar–Uryasev CVaR at level $\alpha=0.95$ and shrink $\hat{\Sigma}_t$ toward the identity (54). The KKT system logs active constraints, tail weights, and duals at each rebalance, the substrate for governance in §4.5.

The agent loop. Algorithm 1 states the plan–act–verify–refine loop of §3 in full. Configurations are code-as-action: the controller may set objective and generator parameters, never weights and never constraint parameters, which the trusted validator owns (§4.5).

Q Full evaluation protocol and metric registry

We evaluate ARC-AGENT under a deliberately broad protocol built to stress nonstationarity across universes and windows; the protocol and synthetic benchmark will be released as a reusable testbed upon acceptance.

What we run, and what we only register. We report measured results on **U3 only**. U3 is a controllable synthetic regime-shift benchmark with *ground-truth latent regimes* and tunable crisis frequency, severity, duration, and detectability. Among the testbeds available to us it is the only contamination-free one: no pretrained model can have memorised its regimes, and the true state is

known, so regime inference can be scored directly and an *oracle bound* can be computed exactly (§5.1). Neither property is available on historical data.

Two historical universes are **preregistered as protocol and deliberately not run**: (U1) a point-in-time cross-asset ETF panel respecting inception dates (TIP 2003, EEM 2003, commodity ETFs 2006+, VXX 2009; a ~ 40 -instrument diversified panel does not exist before the mid-2000s), and (U2) a large-cap equity panel from point-in-time index membership *including delisted names*. We did not execute U1 or U2 in this study, preserving them for a separately preregistered evaluation with independently frozen universe-construction and data-cleaning rules, and no empirical claim in this paper is based on either. App. S records the frozen U1/U2 protocol for transparency.

Horizon and walk-forward (U3, measured). Daily simulated returns over 2,200-day paths, a 756-day warm-up, monthly rebalancing, and 69 evaluated rebalances per run. Estimation is strict, purged, and embargoed walk-forward (53); the expanding walk-forward yields *multiple* OOS sub-periods. Per-instrument transaction costs of 8 bps and an ℓ_1 turnover cap $\|w_t - w_{t-1}\|_1 \leq 0.20$ **per rebalance** are applied identically to all methods. Because U3 is synthetic there is no risk-free series and no calendar: Sharpe is reported on excess return with $rf=0$, and the rf sensitivity of the headline contrast is reported in App. V.

Horizon and walk-forward (U1/U2, registered, not run). Were the historical universes executed, they would use daily total returns over **2007–2025** (the earliest date at which U1 exists as a diversified panel), the same purged and embargoed expanding walk-forward, monthly rebalancing, and three predeclared crisis windows (2008–09, the 2020 COVID crash, and the 2022 inflation shock) as stress slices, with Sharpe taken net of the 3-month T-bill ($rf \approx 1.6\%$ over that span). *None of these settings bears on any number reported in this paper*; they are recorded so that a future execution is bound by them.

Baselines (eleven, all implemented and reported). *Classical* (five): equal weight, risk parity, minimum variance, mean–variance (57), hierarchical risk parity (52). *Robust / scenario* (four) $\rightarrow$ the *same* CVaR-QP, so only scenario quality differs: historical CVaR, DRO-CVaR, a Gaussian scenario model, and a stationary block bootstrap. *Agentic controls* (two): a rule-based weights-emitting agent with no symbolic solve, and the same agent with a projection onto the feasible set. Neither is a language-model agent; no LLM was executed in this work. All allocator-fed methods share ARC-AGENT’s box, turnover cap and costs. Results appear in Table 8 (four), Table 11 (two) and Table 9 (five).

We do not report TimeGAN (68), TimeVAE (38), TSDiff (50), DeepAR (61), or TFT (55). Each is a substantial model, and a half-trained instance would not be a fair comparator. Reporting numbers that were not measured would violate the reporting discipline of this paper; we state the omission rather than fill the row.

Method registry. Table 13 lists *every* method, ablation, oracle and control that appears anywhere in this paper, with the single place each is reported. It exists so that the counts in the text can be audited against the tables: **eleven baselines** (five classical, four scenario-fed, two rule-based controls), one diagnostic comparator, three ARC-AGENT arms, and the oracle and intervention rows. The two control arms are counted once, as baselines; they reappear as comparison rows in Table 11 rather than as additional methods.

Seed and uncertainty registry. Seed counts differ across tables because the experiments were run at different stages of the project, and we report that rather than harmonise it retrospectively. Table 14 states, for every table and figure, the number of seeds, whether the comparison is paired under common random numbers, and what the reported $\pm$ means. We label every experiment **confirmatory** (paired, common random numbers, 10–24 seeds) or **exploratory** (5–6 seeds), and we do not claim that every headline result is confirmatory: the oracle hierarchy and the stage interventions are, while the main baseline table, the red-team table and the 2^3 factorial, including the interaction estimate, are exploratory and should be read as such.

Significance. We assess Sharpe and CVaR differences with a stationary block bootstrap ($R=10^4$ replications; we reserve B for the refinement budget), Diebold–Mariano tests (39) on per-period loss differentials, and the *deflated Sharpe ratio* (53) to correct for selection bias. **Trial counting:**

Table 13: **Method registry.** Every method in the study, its family, whether it is fed by the shared CVaR-QP, and where it is reported. “Not reported” means implemented but **not executed** for this version, so no outcome has been observed; nothing here was run and then withheld.

Method	Family	CVaR-QP	Reported in
Equal weight	baseline (classical)	n/a	Tab. 8
Minimum variance	baseline (classical)	n/a	Tab. 8
Risk parity	baseline (classical)	n/a	Tab. 9
Mean–variance	baseline (classical)	n/a	Tab. 9
Hierarchical risk parity	baseline (classical)	n/a	Tab. 9
Historical CVaR	baseline (scenario)	✓	Tab. 9
DRO-CVaR	baseline (scenario)	✓	Tab. 9
Gaussian scenario model	baseline (scenario)	✓	Tab. 8
Stationary block bootstrap	baseline (scenario)	✓	Tab. 8
Weights-emitting control	baseline (rule-based control)	n/a	Tab. 11
+ projection	baseline (rule-based control)	n/a	Tab. 11
Regime-cond. bootstrap	diagnostic comparator	✓	Tab. 8
ARC-AGENT (full)	proposed	✓	Tab. 8, 11
– verifier	ablation	✓	Tab. 8
– CVaR term	ablation	✓	Tab. 8
– memory	ablation	✓	not reported
– MoE (single expert)	ablation	✓	not reported
$B=1$ (no refinement)	ablation	✓	not reported
L0–L3 oracle hierarchy	oracle (4 rows)	✓	Tab. 10
I0–I2 interventions	intervention (3 rows)	✓	Tab. 5
2^3 factorial arms	intervention (8 cells)	✓	Tab. 6

Table 14: **Seed and uncertainty registry.** “Paired” means the arms share return paths under common random numbers, so the reported interval is on the contrast rather than on the level. Confidence intervals on paired contrasts are 95% percentile intervals from a bootstrap over *seeds* ($R = 10,000$ replications); because time is paired under common random numbers, resampling is over seeds, not over time. The stationary block bootstrap (block length 21 days) is a separate procedure applied to *time-series* Sharpe and CVaR differences (App. Q), and the two should not be conflated. The last column of the table states which experiments are confirmatory and which are exploratory.

Where	Quantity	Seeds	Paired	$\pm$ denotes	Status
Tab. 8	Sharpe by method	5	no	across-seed s.d.	exploratory
Tab. 5	Sharpe by intervention	10	yes	standard error	confirmatory
Tab. 6	Δ Sharpe, channels	6	yes	95% CI	exploratory
Tab. 10	oracle Sharpe	24	yes	point estimates	confirmatory
Tab. 11	Sharpe under attack	5	no	across-seed s.d.	exploratory
Tab. 9	remaining baselines	5	no	across-seed s.d.	exploratory
Fig. 2(a)	oracle frontier	24	yes	± 1 s.e.	confirmatory
Fig. 2(b)	stage interventions	10–16	yes	95% CI	confirmatory
Fig. 2(c)	factorial	6	yes	95% CI	exploratory
Fig. 3	oracle frontier	24	yes	± 1 s.e.	confirmatory
§5.2	perception intervention	16	yes	95% CI	confirmatory
§5.1	L1, L2 paired headroom	24	yes	95% CI	confirmatory
§5.1, App. V	L2 headroom, rf sweep	12	yes	95% CI	confirmatory
§5.3	repaired-generator Sharpe	6	yes	95% CI	exploratory

the effective number of trials is not the handful of rows we report but the full preregistered search: the configuration grid ($K, N, \eta, q, \lambda, \gamma, \alpha, \lambda_\mu$) crossed with seeds, giving the 14,580 enumerated in App. T. Ablations and baselines are *excluded* from that count because they do not search the grid. Deflating against the reported rows would understate the multiplicity; we deflate against the enumerated count and report that a substantial part of our nominal advantage does not survive it.

Metrics. *Portfolio:* CAGR, volatility, **Sharpe net of the risk-free rate** (on U1/U2 the 3-month T-bill; on U3, $rf=0$; reporting excess-return Sharpe matters because a positive rf deflates all Sharpes and differentially flatters low-volatility strategies, i.e. ours, which is why App. V reports the headline contrast at $rf=0$, $rf=1.6\%$ and $rf=3\%$), Sortino (downside deviation below zero excess return), MaxDD, Calmar, tail ratio, and **turnover reported as mean per-rebalance** $\|\Delta w\|_1$ (not annualised), so it is directly comparable to the 0.20 cap. *Scenario calibration:* KS, energy score (ES), variogram score (VS), CRPS (43), VaR unconditional and conditional coverage (51, 36), and CVaR error (bps), all reported as **confidence bounds at the corrected level** of Prop. 2, not as p -values (§4.4). *Agentic-value*, defined precisely:

- **Rejection rate:** fraction of rebalances where the first scenario set fails the certificate.
- **Catch rate:** fraction where refinement was triggered *and* subsequently improved calibration. These are distinct quantities and we report both; conflating them (reporting the rejection rate under the name “catch rate”) would overstate the verifier’s usefulness.
- **Re-planning regret:** $\frac{1}{T} \sum_t [F_t(w_t^{\text{agent}}) - F_t(w_t^{\text{oracle-config}})]$, the mean QP-objective gap between the configuration the controller chose and the best configuration in the grid in hindsight, in objective units. Negative means the controller beat the best *fixed* configuration by adapting.
- **Tool-call budget:** mean tool invocations per rebalance.

We **do not** report *solver-record availability* (elsewhere loosely “provenance completeness”) as a headline metric. It records whether the convex solve emitted a complete active-set and KKT record for w^{sol} , not that the committed w_t is KKT-optimal, and it is 100% by construction for any solver-backed agent and 0% for any that is not, so it carries no information about quality (indeed the —verifier ablation attains 100%; see §6). We report it once, as a property, not as a result.

ARC-AGENT variants and ablations. All executed variants use the **rule-plus-contextual-bandit controller**; *no LLM controller variant was run*, so no rule-versus-LLM comparison appears anywhere in this paper. The preregistered ablation family has five arms, and we state the status of each once, here: —**verifier** (accept the first scenario set) and —**CVaR term** are *executed and reported* in Table 8; —**memory**, —**MoE** (single expert) and $B=1$ (no refinement) are implemented and runnable but were not executed for this version, so no outcome for them has been observed by us or withheld. We record this as a disclosed deviation from the analysis plan: it registered five ablation arms and we report two. No outcome is claimed for the three unexecuted arms anywhere in the paper. This list supersedes any partial enumeration elsewhere.

R Cost, latency, and failure modes

A recurring objection to modular generative pipelines is compute. We quantify it rather than concede it.

Measured cost. The pipeline records wall-clock time per rebalance. Each rebalance performs an HMM posterior update, $N=512$ diffusion samples, verification, and one convex solve; diffusion sampling dominates. The generator is refit every 1,000 trading days, so a run performs 2 refits of 12 epochs each rather than one per rebalance (Table 12; App. O notes that other scripts in the released code use different generator budgets). The entire measured protocol (all *executed* baselines and ablations at five seeds, the red-team suite, the oracle sweep and the figures) reproduces in approximately 25 minutes on a single CPU core, which is the measured compute cost of this study; no GPU is required. A monthly rebalancing interval is $\approx 2.6 \times 10^6$ s, so the solve occupies

$$\frac{0.55 \text{ s}}{2.6 \times 10^6 \text{ s}} \approx 2 \times 10^{-7}$$

of the decision interval, about *one part in five million*, not the $\sim 10^{-5}$ an earlier draft reported. Inference is therefore not compute-bound for the cadence it targets; training, which is excluded from this figure, is the real cost.

Evaluation length. Each U3 run uses a 756-day warm-up followed by 1,444 evaluated trading days, rebalanced monthly (21 trading days), giving 69 rebalances per run. The 2007–2025 calendar, the historical risk-free rate, and the 2008–09/2020/2022 crisis windows belong to the *registered, unexecuted* U1/U2 protocol (App. Q) and bear on no number in this paper.

Inference is negligible; refitting is the dominant computational cost. Within a run, the per-rebalance work (sampling, verification, solve) is small relative to the two generator refits. The 14,580 figure of App. T is an enumerated multiplicity count used to deflate Sharpe; it is not a count of executed runs, and we do not claim to have trained 14,580 generators. What we actually executed is the set recorded in Table 14: twelve method arms at five seeds for Table 8, seven arms at five seeds for the red-team suite, and the oracle and intervention sweeps, which require no generator at all. Deflating against the full preregistered grid is deliberately conservative, and is preferred here to deflating against a count of runs that happened to finish. We report no LLM latency or token figures, because no LLM was executed (App. L). The seed counts we report are a choice we should have made differently; compute was never the constraint. As Table 14 records, they still range from 5 to 24: the oracle hierarchy is paired over 24 seeds and the stage interventions over 10–16, but the main baseline table (Table 8), the red-team table (Table 11) and the factorial (Table 6) rest on 5, 5 and 6 seeds respectively. We treat the 24- and 10–16-seed paired contrasts as confirmatory and the 5- and 6-seed tables as exploratory, and every claim drawn from the latter is labelled accordingly rather than argued around.

S Universe construction (registered, not run)

U1 would be built point-in-time: an instrument enters only after its inception date and only if it meets a liquidity floor at t , so the panel *grows* from ~ 14 instruments (2007) to ~ 40 (2015 onward), and d is time-varying. U2 would use point-in-time index membership including subsequently delisted names. **Neither was run.** We register the construction so that a future execution is bound by it, and so that our own claims cannot quietly expand to cover data we never touched.

T Enumerated trial count

Deflating a Sharpe ratio requires the number of *trials*, not the number of reported rows. Our frozen preregistration enumerates the grid $K \in \{2, 3, 4\}$, $N \in \{256, 512\}$, $\eta \in \{1, 2, 3\}$, $q \in \{0.05, 0.10\}$, $\lambda \in \{0.3, 0.5, 0.7\}$, $\gamma \in \{0.5, 1.0, 1.5\}$, $\alpha \in \{0.90, 0.95, 0.99\}$ and $\lambda_\mu \in \{4, 12, 24\}$. The Cartesian product is $3 \cdot 2 \cdot 3 \cdot 2 \cdot 3 \cdot 3 \cdot 3 \cdot 3 = 2,916$ configurations, and with the five registered seeds this is 14,580 grid-by-seed combinations, the count passed to the deflated-Sharpe calculation. This is an enumerated multiplicity count, not a tally of executed runs, and it is used in that sense everywhere in the paper. No configuration is pruned. The preregistration also records an upper bound that treats each *combination* of ablation arm and controller arm as a separate search: five ablation arms $\times$ two controller arms = 10 families, giving $10 \times 14,580 = 145,800$. We deflate against the core 14,580, not against that bound, because ablations and baselines do not search this grid; the upper bound is recorded only so that a reader can see the most adverse reading. Deflating against the eight rows of Table 8 would understate selection bias by three orders of magnitude, and is a common and consequential error. At this trial count none of the performance margins in Table 8 survive deflation, which is consistent with, and reinforces, our headline negative result.

Preregistration amendment (disclosed). One amendment (PR1) was recorded after freezing and before any hypothesis outcome was inspected: the range of λ_μ was widened from $[0, 2]$ to $[0, 30]$. Reason: at a daily horizon the CVaR term ($\sim 10^{-2}$) is $\sim 33\times$ the return term ($\sim 3 \times 10^{-4}$), so the original bound made the return/risk trade-off *inexpressible*: the objective degenerated to pure CVaR minimisation and the agent performed no regime rotation at all. This is a specification bug; the preregistration was re-frozen with a new hash. The amendment, the hashes and the audit scripts will be released upon acceptance.

Table 15: **Interventional attribution, measured.** Shapley values and Shapley–Taylor interaction indices computed from the complete 2^3 coalition table (Table 6; 6 seeds, paired, 95% CI by seed bootstrap). **The two blocks are alternative attribution summaries, not additive pieces of one decomposition:** block A already sums to $v(N) = +0.218$ exactly, as efficiency requires, and block B is a further diagnostic of non-additivity that must not be added on top of it. Covariance is inert with tight intervals throughout. The mean $\times$ scenario interaction is 55% of $v(N)$ as a point estimate and the *total* pairwise interaction share is 54%; both are underpowered at six seeds. These are descriptive attribution summaries: neither the interaction magnitude nor the conditional contribution of the mean channel is established by them.

	ϕ	95% CI	share of $v(N)$
<i>A. Component credit</i> (Shapley values; sum to $v(N)$)			
conditional mean $\hat{\mu}$	+0.080	± 0.178	36.8%
scenario tail/dependence	+0.139	± 0.086	63.8%
covariance $\hat{\Sigma}$	−0.001	± 0.002	−0.6%
<i>B. Interaction diagnostics</i> (Shapley–Taylor indices; not added to A)			
mean $\times$ scenarios	+0.120	± 0.184	55.3%
mean $\times$ covariance	−0.002	± 0.002	−0.9%
covariance $\times$ scenarios	−0.001	± 0.003	−0.3%
higher-order (three-way)	−0.003	n/a	−1.5%
total pairwise interaction share		54.1% of $v(N)$	

U Attribution: full coalition table

Efficiency ($\sum_i \phi_i = v(N)$) holds without approximation here because the coalition table is complete: no Monte-Carlo sampling over permutations is needed for three modules. That exactness is combinatorial rather than statistical: each $v(S)$ is a six-seed paired estimate, so the ϕ_i inherit six-seed uncertainty, which is what the intervals in Table 15 report. The same computation on an incomplete table would additionally require estimating $v(S)$ for unrun coalitions, which is exactly the step at which attribution studies usually become assumption-driven.

Two summaries, not one decomposition. Block A of Table 15 partitions $v(N)$ among the three channels. Block B measures how much of $v(N)$ is attributable to *non-additivity* between them. A channel’s Shapley value already absorbs its share of the interactions it participates in, so the two blocks describe the same coalition table from two directions and cannot be concatenated into a single bar chart of additive shares. Figure 2(c) plots both, and its caption says so.

V Representation of information and action

§5.1 varies the action set at fixed information. The general object is a lattice indexed by an information set I and an action set A , with $J^*(I, A) = \sup_{\pi \in \Pi(I, A)} J(\pi)$. Our L0–L3 hierarchy is the column $I = \{\text{true regime path}\}$ of that lattice.

Proposition 1 (Lattice monotonicity). *If $I_1 \subseteq I_2$ and $A_1 \subseteq A_2$ then $J^*(I_1, A_1) \leq J^*(I_2, A_2)$.*

Proof. Any $\pi \in \Pi(I_1, A_1)$ is measurable with respect to I_1 , hence with respect to $I_2 \supseteq I_1$, and takes values in $A_1 \subseteq A_2$; so $\Pi(I_1, A_1) \subseteq \Pi(I_2, A_2)$ and the supremum over the larger set is at least as large. The argument requires only that additional information and additional actions may be *ignored*, which holds because Π is defined by measurability and range constraints alone. $\square$

The proposition itself is immediate; the content is its experimental implication. A low headroom estimate in one information–action cell cannot establish that the information has little value, and a rise in headroom when the action set is widened at fixed information shows that the earlier negative conclusion came from representational restriction. §5.1 is exactly that situation: L1 gave $+0.07 \pm 0.08$ and L2 gave $+0.28 \pm 0.08$ on identical information. Reporting a single oracle number reports one cell of this lattice without saying which.

The three headroom numbers, side by side. Three quantities describe the L2 headroom and are easy to read as inconsistent, so we state each with its provenance. (a) The **paired same-path Sharpe difference** L2–L0 is $+0.28 \pm 0.08$ (24 seeds, common random numbers, 95% CI over seeds), computed, like every level in Table 10, at the preregistered $rf=1.6\%$. (b) The same contrast in the twelve-seed rf sweep (seed registry, Table 14) is $+0.248$ at $rf=0$, $+0.263$ at $rf=1.6\%$ and $+0.275$ at $rf=3\%$; §5.1 quotes the first of these as $+0.248$, while the abstract and Table 10 quote (a). (a) and (b) differ in seed count and in rf , not in what they measure, and neither is a correction of the other. (c) $+0.38$ is not a paired difference against L0 at all: it is the *vertical* excess of L2 over the static EW–MinVar frontier *at matched drawdown*, the “Excess” column of Table 10 and the gap drawn in Fig. 2(a). (a) and (b) compare two policies on the same paths; (c) compares one policy against a frontier at its own realised risk. A reader who subtracts (c) from (a) is subtracting two different comparisons.

W Conditional validity and unidentified steps

We label this appendix *justification* rather than *proofs*: it composes three standard results under Assumption 1 and states explicitly, at the end, the one step we have not proved.

Assumption 1 (Conditions for the conditional certificate statement 1). (C1) *Bounded loss.* The portfolio loss $-w^\top r$ is bounded by M uniformly over the feasible set of (3), and the N scenarios are drawn i.i.d. from $p_\theta(\cdot \mid z_t)$ given z_t . (C2) *Strong convexity.* The objective (5) is Γ -strongly convex in w on the feasible set, with moment sensitivities $(\lambda_\mu, \kappa_\Sigma)$ and moment errors $(\delta_\mu, \delta_\Sigma)$ as in Lemma 1. (C3) *Score construction and approximate local exchangeability.* The calibration score s_t is the absolute realised-versus-scenario CVaR discrepancy of Definition 1, computed from r_{t+1} after w_t is committed, so it is not used to select w_t ; the score sequence is approximately locally exchangeable in the sense adaptive conformal inference is designed to tolerate (42, 69). We do not verify this empirically. (C4) *Selection correction.* At most B refinement attempts occur per rebalance and each attempt is evaluated at level δ/B (Prop. 2). (C5) *Score-to-CVaR identification.* There exists a monotone link between the calibrated quantiles of $\{s_\tau\}$ and the population discrepancy $|\text{CVaR}_\alpha^{p_\theta} - \text{CVaR}_\alpha^\mathbb{P}|$, so that coverage of the former bounds the latter. **We do not prove this**, and everything below is conditional on it. (C6) *Per-attempt marginal validity.* Each candidate attempt satisfies the stated marginal validity condition *before* adaptive selection, so that the union bound of (C4) applies. **We do not establish this** for the implemented proposal mechanism.

Under (C1)–(C6), which include the unproved score-to-CVaR identification (C5) and per-attempt validity (C6), the certificate below is a **long-run** statement: the guarantee is over the empirical frequency of covered rebalances, never over a nominated one. We use “per-rebalance” only to describe the *form* of the bound, never its validity.

Three properties follow. (i) The verifier is load-bearing: q_t is computed *from the verifier’s realized scores*, so the bound cannot even be evaluated without running it. (ii) It bounds model error, by calibrating against $\mathbb{P}$ -outcomes rather than p_θ -samples. (iii) The refinement correction does real work.

The body summarises the conditional certificate statement 1 and Prop. 2 without proof; the full development follows.

The allocator’s stability under objective perturbation is standard; the substantive question is what the verifier can certify about the gap between the optimised objective and the objective under the *true* distribution. We first isolate that gap, then show why a Monte Carlo bound cannot close it, then close it with a conformal certificate calibrated on realized outcomes.

The two error terms. Scenarios are drawn from the generator $p_\theta(\cdot \mid z_t)$, not from the true law $\mathbb{P}$, so the CVaR error that the allocator’s regret depends on decomposes into a sampling term (S) and a model term (M). Term (S) shrinks with more scenarios; term (M) is the generator’s *miscalibration* and does not shrink with N at all. Term (M) is the entire reason the verifier exists, and it is the one a concentration inequality on p_θ -samples cannot touch: such a bound certifies only that we have sampled our own generator faithfully.

Lemma 1 (Allocator stability). *If (5) is Γ -strongly convex in w and its moment and CVaR terms differ from truth by $(\delta_\mu, \delta_\Sigma, \delta_{\text{CVaR}})$, then $F(\hat{w}) - F(w^*) \leq \frac{1}{2\Gamma}(\lambda_\mu \delta_\mu + \kappa_\Sigma \delta_\Sigma + \delta_{\text{CVaR}})^2$. This is standard; the content is what bounds δ_{CVaR} , below.*

Lemma 2 (Sampling error, term (S)). *For the bounded linear loss class $\{-w^\top r : \|w\|_1 \leq C\}$ with $|L| \leq M$, a Rademacher bound for the empirical tail average (34, 65, 63) gives, with probability $\geq 1 - \delta$, uniformly over feasible w , $\sup_w |\widehat{\text{CVaR}}_\alpha^{p_\theta} - \text{CVaR}_\alpha^{p_\theta}| \leq \frac{cM}{1-\alpha} \sqrt{\frac{\log d + \log(2/\delta)}{N}}$.*

The $\sqrt{\log d}$ rate (not $\sqrt{d \log N}$) is the correct one for a linear class under an ℓ_1 budget. This term is real but secondary; the interesting term is (M), and it requires *realized* data.

Model error requires realized outcomes. Within the implemented gate and the available data, the only direct empirical evidence about $\text{CVaR}^\mathbb{P}$ is the sequence of realized returns r_{t+1} actually observed after each decision. We therefore build the verifier as a *conformal-style risk controller*: at each rebalance it scores the realized outcome against the scenario predictive distribution and calibrates the certificate width from the empirical distribution of past scores, rather than asserting a bound analytically.

Definition 1 (Realised calibration score). Fix $\alpha = 0.95$. Let $\ell_{t+1} = -w_t^\top r_{t+1}$ be the realised portfolio loss and let $\widehat{\text{CVaR}}_\alpha$ be the CVaR of the *accepted* scenario set, computed as the mean of those scenario losses at or above their own empirical α -quantile. The score is the **absolute** one-step discrepancy

$$s_t = \left| \ell_{t+1} - \widehat{\text{CVaR}}_\alpha \right|,$$

in units of daily return. It is a **nonnegative absolute discrepancy** by construction, is computed *after* w_t is committed, and is the only formula used anywhere in this paper: the verifier maintains the rolling window $\{s_\tau\}_{\tau < t}$ of the most recent 60 scores and takes its empirical quantile.

Conditional certificate statement 1 (Act-time decision bound, conditional on the unproved (C5) score-to-CVaR identification and (C6) per-attempt validity). *Assume the score-to-CVaR identification condition and the per-attempt validity condition of Assumption 1, neither of which is proved here. Run adaptive conformal inference (42, 69) on $\{s_\tau\}$ with target level β_t updated online, and gate scenarios on the resulting interval. Then the model-error term satisfies, with long-run frequency $1 - \delta'$ under the approximate local exchangeability of Assumption 1(C3), $|\text{CVaR}_\alpha^{p_\theta} - \text{CVaR}_\alpha^\mathbb{P}| \leq q_t(\delta')$, where $q_t(\delta')$ is the adaptively calibrated width at the per-attempt level $\delta' = \delta/B$. Combining with Lemma 2 and the allocator stability lemma, the committed decision obeys, on the covered rebalances and with long-run frequency $1 - \delta$,*

$$F(\hat{w}_t) - F(w_t^*) \leq \frac{1}{2\Gamma} \left(\underbrace{\lambda_\mu \delta_\mu + \kappa_\Sigma \delta_\Sigma}_{\text{model, conformal}} + \underbrace{q_t\left(\frac{\delta}{B}\right)}_{\text{sampling, Lem. 2}} + \underbrace{\varepsilon_N\left(\frac{\delta}{B}\right)}_{\text{sampling, Lem. 2}} \right)^2. \quad (6)$$

Correction (supersedes an earlier draft). A previous version asserted that an unbounded refinement loop “admits no nontrivial certificate at any level.” **That is false, and we withdraw it.** What the union bound shows is that *Bonferroni* over B attempts degrades to zero as $B \rightarrow \infty$, a statement about that proof technique, not about certifiability. Anytime-valid inference (e-processes, Ville’s inequality, always-valid p -values) is designed precisely to license optional stopping and *does* certify unbounded refine-until-pass loops. The correct claim is therefore: naive refinement invalidates the certificate; a union bound repairs it for bounded B assuming the per-attempt validity condition (C6); anytime-valid inference is the principled route to unbounded B , and we do not take it.

Proposition 2 (Refinement is selection; Bonferroni restores validity for bounded B , assuming (C6)). Assuming each candidate attempt satisfies the stated marginal validity condition before adaptive selection (*a condition we do not establish for the implemented proposal mechanism*), then under $\leq B$ refinement attempts, evaluating the conformal interval at level δ/B per attempt yields a certificate valid for the accepted attempt despite selection on s_t . As $B \rightarrow \infty$ the Bonferroni-corrected level degrades to zero, so this proof technique yields no nontrivial certificate in the limit; anytime-valid constructions certify unbounded refine-until-pass loops, and we do not take that route here.

Nonstationarity is why it must be adaptive conformal. Standard conformal assumes exchangeability, which fails precisely at regime change, exactly when the certificate is most needed. Adaptive conformal inference (42) targets long-run coverage by updating β_t from realized miscoverage, which is why we adopt it rather than a fixed split-conformal bound. We say “targets” rather than “guarantees under arbitrary shift”: the guarantee we invoke holds under the approximate local exchangeability assumed in Assumption 1(C3), which we do not verify empirically.

Proposition 3 (Audited action admissibility and solver-output provenance). *Provided (ℓ, u, τ) come from the trusted validator, the convex program enforces the budget, box and turnover constraints on its solver output w^{sol} , so an erroneous controller cannot enlarge the feasible set. In all reported executions the post-processed committed vector w_t **passed the specified post-hoc feasibility audit** at the tolerances of App. O, which also gives the validate-and-reject variant that would make this a guarantee rather than an audit. The logged active set and KKT duals constitute **optimality evidence for w^{sol}** ; for w_t we claim only **empirically audited admissibility**.*

Corollary 1 (Empirically audited feasibility and solver provenance). *Independently of controller quality, generator quality, or whether the certificate holds, every committed action in the reported experiments passed the budget, box and turnover checks at the tolerances of App. O. Separately, the solve behind each action carries a reproducible KKT record for the solver output w^{sol} (Prop. 3). These are empirical properties of the reported executions and of the logged solver outputs; they are not unconditional guarantees for arbitrary future executions, nor optimality certificates for the post-processed committed vector. Decision-quality certification, feasibility auditing and solver provenance are distinct properties and we report them separately.*

Remark 1 (What the certificate assumes). The act-time bound rests on (a) bounded losses and i.i.d. sampling from p_θ (term S), and (b) approximate local exchangeability of calibration scores, which adaptive conformal is designed to tolerate under shift (term M). It does *not* assume $p_\theta \approx \mathbb{P}$; that gap is measured, not assumed. This is the substantive difference from a goodness-of-fit gate, which asserts calibration it cannot observe.

What we have not proved. One step in the chain is asserted rather than demonstrated, and we flag it here. Adaptive conformal inference gives long-run coverage for a *future calibration score* s_t . It does not, without further argument, yield a bound on the distance between two *population* functionals, $|\text{CVaR}_\alpha^{p_\theta} - \text{CVaR}_\alpha^{\mathbb{P}}| \leq q_t(\delta')$. Closing that gap requires (i) a precise statement of how the score of Definition 1 identifies the population CVaR difference, in particular a monotone link between score quantiles and that difference, and (ii) an argument that each adaptively proposed refinement attempt is individually valid at level δ/B *before* the union bound is applied over attempts. We have neither, and statement 1 should be read as conditional on that link, which is why it is not labelled a theorem.

Two further limits follow from the construction rather than from any gap in it. The coverage is long-run and marginal, so no individual rebalance is certified. And (C3) is an assumption about the score sequence that we do not verify empirically here. H2 was never exercised in our runs (§5.5), so nothing in this paper tests the refinement correction. Accordingly, the honest reading of the verifier in this paper is as an empirically motivated safeguard with a stated but incomplete theoretical basis, and a reader who prefers to discount the certificate entirely loses none of the empirical results: every measured claim in §5 is independent of it, and feasibility and provenance rest on Prop. 3 alone.